

The most probable first-passage time of the lazy walk in a reflecting box:

rigorous results, a dependency-ordered synthesis

Companion note R0

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Note. This document is the companion note R0 of the article “Mean and most probable first-passage times of lazy random walks in reflecting hypercubic lattices” ([paper.pdf](#), Section 7; Supplementary Material, Section S8), in the repository <https://github.com/zhouyi-xiaoxiao/master-dissertation>. In the repository, file paths are relative to the root of the repository; the run logs cited below are in `data/msc_rigorous_*/logs/` and `data/checks/`, the code of the internal checks is not part of the repository, and `notes/VERIFICATION.md` summarises how the notes were checked.

Abstract

For the lazy nearest-neighbour walk on $\{1, \dots, N\}^d$ with move probability q , reflecting (move-cancelling) walls and one absorbing target site, we collect in one dependency-ordered document the results on the first-passage law and on its mode t^* that are proved in five companion notes, with unified notation and with the status of every statement stated explicitly. In one dimension the scaled mode converges, $qt^*/(N - \frac{1}{2})^2 \rightarrow c = 0.3332842654949478987\dots$, with explicit finite- N windows for every $N \geq 3$ when $q \leq \frac{4}{5}$; there the two-pole closed form $t_{\text{cf}} = \mathbb{E}T \ln(2dX)/(X + 2d - 1)$, $X = \mu_1 \mathbb{E}T$, is *false* as an approximation (on the certified ranges, $10 \leq N \leq 1500$ and $N = 2000$ for $q = \frac{4}{5}$ and $10 \leq N \leq 1000$ for $q = \frac{1}{2}$, it overestimates the mode by more than 15.7%). In two and three dimensions, for opposite corners, the closed form is a computer-assisted theorem with an explicit error: $X\mu_1(t^* - t_{\text{cf}})$ lies in a certified bounded window for every $N \geq N_0$ (continuous time; discrete time for $q \leq \frac{1}{2}$ and for $q \leq 0.99$; for every $q < 1$ when $N \geq \max(N_*, N_1(q))$), which yields the laws $t^* = \frac{4N^2}{\pi^2 q} [\ln \ln N + \ln 8\pi + O(\frac{\ln \ln N}{\ln N})]$ ($d = 2$) and $t^* = \frac{6N^2}{\pi^2 q} [\ln N + 3.7528\dots + O(\frac{\ln N}{N})]$ ($d = 3$) with explicit remainders; on finite ranges the relative error of the closed form is certified to be below 0.971% ($d = 2$) and 0.275% ($d = 3$). Unimodality of the first-passage law of the three geometries is a theorem in continuous time for all N and d , and a computer-assisted theorem in discrete time for every $q \leq \frac{4}{5}$ (sharp; corner to corner in every dimension, the two centre geometries for $d \geq 2$); it fails for general start–target pairs and, in discrete time, near $q = 1$. The spectral structure (poles, interlacing, residues, explicit bounds, the constants C_2 , C_3 of the mean first-passage time) is proved for every N (the numerical values of the constants and of the explicit remainders with computer assistance). We list the certificates with their trusted bases and the commands to re-verify them, report a serial re-run of all sanity scripts, and list precisely what remains open.

Contents

1	How this document is organised	2
2	Setting and notation	4
3	Summary table of all statements	5
4	Results for every finite symmetric chain and every start–target pair	7
4.1	Spectral representation, interlacing, residues	7
4.2	Reflection-symmetric pairs: the signs of the residues	8
4.3	Laziness monotonicity	9
4.4	The tail lemma (cone certificate)	10
4.5	Last-exit decomposition and a criterion for unimodality	10
4.6	Explicit bounds on the principal pole and the neglected poles	10

5	One dimension	11
5.1	The exact law	11
5.2	Unimodality and log-concavity	11
5.3	Starts away from the reflecting wall	14
5.4	The continuum limit and the constant c	14
5.5	The mode	15
5.6	General start	16
5.7	Certified modes on finite ranges	16
6	Unimodality in every dimension	17
6.1	The three geometries for all N and d	17
6.2	Discrete time up to the value $q = \frac{4}{5}$	17
6.3	Beyond $\frac{4}{5}$, and $q = 1$	18
6.4	Explicit windows for the CC pair, every q	18
6.5	General pairs	19
6.6	Certified modes in two and three dimensions	19
7	Spectral quantities of the corner-to-corner box	19
7.1	Every dimension	19
7.2	Two dimensions	21
7.3	Three dimensions	22
7.4	Consequences for the poles	22
8	The mode in two and three dimensions and the closed form	23
8.1	Decomposition and unimodality	23
8.2	An exact identity and the window theorem	23
8.3	The closed form in continuous time	24
8.4	Discrete time	24
8.5	Certified accuracy on finite ranges, and a combination	25
9	Certificates: trusted bases and how to re-verify them	26
10	Re-run of the sanity checks	28
11	Internal checks	30
12	What remains open	30
	References	31

1 How this document is organised

Purpose. The article studies the most probable first-passage time (the *mode*) of a lazy random walk in a box to a single target site, and proposes the closed form

$$t^* \approx t_{\text{cf}} := \mathbb{E}T \cdot \frac{\ln(2dX)}{X + 2d - 1}, \quad X := \mu_1 \mathbb{E}T, \quad \mu_1 := \frac{q}{d} \left(1 - \cos \frac{\pi}{N}\right),$$

which in the article is obtained by a formal two-pole argument. The results below replace the formal argument by theorems where this has been possible, and state precisely where it has not.

The companion notes. The proofs are contained in five companion notes, the companion notes R1–R5 in the folder `proofs/` of the repository. They are cited as follows (statement numbers are those of the PDF of each note).

- 1D *The most probable first-passage time of the lazy walk on a segment: a rigorous one-dimensional theory* (`proofs/R1_one_dimension/`).
- SP *Rigorous spectral structure of first-passage times to a point target on the reflecting hypercubic lattice* (`proofs/R2_spectral_structure/`).
- AS *The mode of the corner-to-corner first-passage time in two and three dimensions: rigorous asymptotics* (`proofs/R3_mode_two_three_dimensions/`).

UN *Unimodality of point-target first-passage laws on the reflecting box: theorems, counter-examples, certified computations, and what remains open* (`proofs/R4_unimodality/`).

CE *Computer-assisted certification of first-passage modes, unimodality, the closed-form mode formula and two constants for the lazy walk in a reflecting box* (`proofs/R5_certified_computations/`).

A reference such as [1D, Thm 6.3] means Theorem 6.3 of the note 1D. File paths in this document are relative to the root of the repository. This document is self-contained as far as statements, definitions and the logical order are concerned. Short proofs are reproduced in full, among them most of the proofs that were written after the internal checks (marker [†] below); long proofs and computer-assisted proofs are given as outlines with an exact pointer to the complete proof in the companion note.

Status of statements. Every numbered statement other than a definition or a remark carries one of the following labels, and no other. (A remark either comments or asserts something that is proved in place; the status of each asserting remark is given in the table of Section 3.)

- **Theorem / Lemma / Proposition / Corollary:** a complete proof is given here or in the cited note, with standard results cited by author, title and theorem number. A *Proposition* may also be a statement proved under an explicitly stated hypothesis that is not itself proved; this is then said in the statement.
- **Computer-assisted theorem** (and lemma, proposition, corollary): the proof reduces the statement, by lemmas proved by hand, to finitely many inequalities that are verified by a program in exact integer or rational arithmetic or in rigorous ball/interval arithmetic. The trusted base and the certificate files are named (Section 9).
- **Conjecture:** not proved; the evidence is stated.
- **Numerical observation:** a reproducible computation, exact or in floating point as stated, with no claim beyond the computed range.
- **Refuted statement:** a natural conjecture, suggested by the one-dimensional case, by a floating-point study or by an intermediate step of a proof, that is false; the counter-example or disproof is given.

Two markers record proof provenance; theorem and conjecture labels state the mathematical status.

[†] A proof developed after the companion-note checks, reproduced here when short or located by its companion-note reference.

[‡] A deduction combining results from different companion notes; the combining argument is given here.

Internal checks. Each companion note was examined by two internal checks carried out within the same project and run on the same machine: a line-by-line reading of every proof (with re-derivation of the displayed identities, a check of every citation and a re-run of the certificate scripts on a private copy), and a numerical search for counter-examples with separately written code that tests every statement with its explicit constants. These checks are not external peer review, and the re-computations in them are not an external replication. Their outcome is summarised in Section 11 and in `notes/VERIFICATION.md`, Section 3. No statement is given a stronger status here than the one supported by these checks.

Decimals. A decimal expansion followed by dots (0.3332...) is truncated: every digit shown is a digit of the exact value. Decimals that appear in inequalities are exact rational numbers. Every quantitative statement of the L^AT_EX source of this document is preceded by a comment `% src:` naming the certificate, data or log file behind it (paths relative to the root of the repository; the cited run logs are in `data/msc_rigorous_*/logs/` and `data/checks/`).

Dependency order. Section 2 fixes the notation. Section 4 contains the results that hold for every finite symmetric chain and every start–target pair; they use nothing else. Section 5 (one dimension) uses Section 4 only, except part (c) of Computer-assisted theorem 5.10, which for odd $N \geq 25$ uses Computer-assisted theorem 6.3 of Section 6 (that theorem does not use Section 5, so there is no circularity). Section 6 (unimodality in every dimension) uses Sections 4 and 5. Section 7 (spectral quantities of the box) uses Section 4. Section 8 (the mode in $d = 2, 3$ and the closed form) uses Sections 4 and 6, and its finite-range results (Computer-assisted theorem 8.14 and Computer-assisted corollary 8.15) use in addition the elementary rational enclosure of the mean first-passage time, Lemma 7.3 of Section 7; it does *not* use the estimates of Section 7, whose relation to it is explained in Remark 8.6. Sections 9–12 contain the certificates, the re-run of the sanity checks, the outcome of the internal checks and the open problems. Section 3 is a table of all statements.

2 Setting and notation

Definition 2.1 (the walk). Let $d \geq 1$, $N \geq 2$, $\Lambda = \{1, \dots, N\}^d$, $n = N^d$. For $x \in \Lambda$ let $b(x)$ be the number of the $2d$ unit vectors $\pm e_i$ with $x \pm e_i \notin \Lambda$. The walk with activity $q = 1$ has the symmetric stochastic matrix $P_1(x, x \pm e_i) = \frac{1}{2d}$ if $x \pm e_i \in \Lambda$, $P_1(x, x) = \frac{b(x)}{2d}$; put $\mathcal{L} := I - P_1$. For $q \in (0, 1]$ the *lazy walk* has transition matrix $P_q := I - q\mathcal{L}$: each of the $2d$ directions is attempted with probability $\frac{q}{2d}$, the walk rests with probability $1 - q$, and an attempted move that would leave Λ is cancelled. The *continuous-time walk* with rate q has generator $-q\mathcal{L}$ (each allowed move at rate $\frac{q}{2d}$). Fix a target $a \in \Lambda$ and a start $x_0 \neq a$; put $\Lambda' = \Lambda \setminus \{a\}$, let $\mathcal{L}_a$ be the principal submatrix of $\mathcal{L}$ on Λ' (the killed generator) and $Q_q := I - q\mathcal{L}_a$. Let T be the first time $t \geq 1$ (resp. $t > 0$) at which the walk started at x_0 is at a , and

$$f_q(t) := \mathbb{P}(T = t) = q e_{x_0}^\top Q_q^{t-1} r \quad (t \geq 1), \quad g_q(t) := q e_{x_0}^\top e^{-qt\mathcal{L}_a} r \quad (t \geq 0), \quad r := \mathcal{L}_a \mathbf{1},$$

the first-passage probability mass function (PMF) and the continuous-time density; $r(x) = P_1(x, a)$. $D := |x_0 - a|_1$ is the graph distance. A *mode* is a maximiser of f_q over $t \geq 1$ (of g_q over $t \geq 0$); t^* denotes any mode.

Definition 2.2 (shape). A nonnegative sequence $(a(t))_{t \geq 1}$ is *unimodal* if there is m with $a(1) \leq \dots \leq a(m) \geq a(m+1) \geq \dots$; *strictly unimodal* if it is zero up to some $t_0 - 1$, strictly increasing on $[t_0 - 1, m]$ and strictly decreasing from m on (then m is the unique maximiser); *log-concave* if its support is a set of consecutive integers and $a(t)^2 \geq a(t-1)a(t+1)$ for all t . A C^1 function $g \geq 0$ on $[0, \infty)$ is unimodal if there is $m \geq 0$ with $g' \geq 0$ on $(0, m)$ and $g' \leq 0$ on (m, ∞) .

Geometries. *Corner to corner* (CC): $x_0 = (1, \dots, 1)$, $a = (N, \dots, N)$, $D = d(N - 1)$. For odd N and $c = \frac{N+1}{2}$: *corner to centre* (C2M): $x_0 = (1, \dots, 1)$, $a = (c, \dots, c)$; *centre to corner* (M2C): $x_0 = (c, \dots, c)$, $a = (N, \dots, N)$. These are the three geometries of the article. In $d = 1$ the CC pair is the walk on the segment $\{1, \dots, N\}$ from the reflecting end 1 to the absorbing end N ; for a general start x_0 on the segment we write $d_0 := N - x_0$ and $\xi_N := (x_0 - \frac{1}{2})/(N - \frac{1}{2})$.

Spectrum of the box. The eigenvalues of $\mathcal{L}$ are $\lambda_k = \frac{1}{d} \sum_{i=1}^d (1 - \cos \frac{\pi k_i}{N})$, $k \in \{0, \dots, N - 1\}^d$, with the orthonormal eigenvectors $\phi_k(x) = \prod_i \gamma_{k_i} \cos(\frac{\pi k_i}{N} (x_i - \frac{1}{2}))$, $\gamma_0 = N^{-1/2}$, $\gamma_j = (2/N)^{1/2}$ [SP, Lemma 1.1], [UN, Lemma 1.1], [AS, Lemma 1.1]. We write

$$\Lambda_1 := \frac{1}{d} (1 - \cos \frac{\pi}{N}) = \frac{2}{d} \sin^2 \frac{\pi}{2N}, \quad \mu_1 := q\Lambda_1, \quad W := 2d \cos^2 \frac{\pi}{2N}.$$

Λ_1 is the smallest non-zero eigenvalue of $\mathcal{L}$ (the relaxation rate of the slowest mode), and W is its weight at a corner, $W = n \sum_i \phi_{e_i}(a)^2$.

Mean times and the closed form. For $x \in \Lambda$ let $m(x)$ be the mean hitting time of a from x for the continuous-time walk with rate 1; for activity q the mean first-passage time is $\mathbb{E}_x T = m(x)/q$ in continuous and in discrete time [SP, Lemma 6.1]. Let $\tau := \frac{1}{n} \sum_{x \in \Lambda} m(x)$ be the mean hitting time from the uniform (stationary) start, and

$$X := \mu_1 \mathbb{E}_{x_0} T = \Lambda_1 m(x_0), \quad \bar{X} := \Lambda_1 \tau, \quad t_{\text{cf}} := \mathbb{E}_{x_0} T \cdot \frac{\ln(2dX)}{X + 2d - 1}, \quad \tau_{\text{cf}} := \mu_1 t_{\text{cf}} = \frac{X \ln(2dX)}{X + 2d - 1}.$$

X and $\bar{X}$ do not depend on q ; $\tau^* := \mu_1 t^*$ is the *scaled mode*. For the segment ($d = 1$, CC) we use $L := N - \frac{1}{2}$ and $\mathbb{E}T = N(N - 1)/q = (L^2 - \frac{1}{4})/q$.

Two thresholds. All eigenvalues of P_q are ≥ 0 iff $q \leq q_N^\square := 1/(1 + \cos \frac{\pi}{N})$. For the segment with one absorbing end all eigenvalues of Q_q are ≥ 0 iff $q \leq q_N^\downarrow := 1/(2 \cos^2 \frac{\pi}{2N-1})$. Both are $> \frac{1}{2}$ and decrease to $\frac{1}{2}$; $q_N^\downarrow > q_N^\square$ [1D, (5)], [UN, Thm 4.12].

Dictionary. The notes use partly different symbols; the translation is:

this document	notes	remark
$\mathcal{L}, \mathcal{L}_a$	L, L_a (SP); L, A (UN)	AS uses $\frac{q}{d} \sum_i A_i$
Λ_1	Λ_1 (SP); β_1 (UN)	
$\mu_1 = q\Lambda_1$	μ (AS); μ_1 (CE, article)	
$\bar{X} = \Lambda_1 \tau$	$\bar{X}$ (SP); $X_s = \mu \mathbb{E}_\pi T$ (AS)	
$X = \Lambda_1 m(x_0)$	X (AS, SP Thm 11.3); X_N (CE)	
$m(x_0)$	m (SP); $\tau_N = q \mathbb{E} T$ (CE)	
$\tau^* = \mu_1 t^*$	τ^* (AS)	scaled time
$c = 0.33328 \dots$	c (1D); $\kappa_1 = 2\tau^*$ (CE)	one-dimensional constant
$q_N^ $	q_N (1D)	segment
$q_N^\square$	q_N (UN)	box
residues r_j , poles σ_j	r_j, σ_j (SP); $C_j = r_j \sigma_j$, α_j (UN); $\rho_j \hat{\varphi}(-\nu_j)$, ν_j (AS)	unit rate
$\Lambda = \{1, \dots, N\}^d$	$\Omega = \{0, \dots, N-1\}^d$ (CE)	coordinates shifted by one

In Sections 4 and 7 we write $\varkappa := K_2/\tau^2$ for the quantity called κ in [SP, (6.3)]; the letter κ is reserved for the one-dimensional constant of Theorem 5.17.

3 Summary table of all statements

Status: **T** theorem (lemma, proposition, corollary with complete proof), **CA** computer-assisted theorem, **P** proposition under an unproved hypothesis, **C** conjecture, **O** numerical observation, **F** refuted (false); markers $^\dagger, \ddagger$ as in Section 1. “Here” is the statement in this document, “source” the complete proof.

statement	status	here	source
<i>Every finite symmetric chain, every pair</i>			
poles = zeros of the secular function, interlacing, spectral representation	T	4.1, 4.2	SP 2.2, 3.1, 4.1, 4.2
residues: sum 1, $r_0 > 0$, $\geq D-1$ sign changes, stationary and quasi-stationary starts	T	4.3	SP 5.1, 5.4, 5.5; UN 3.2
single start at distance ≥ 2 is not a mixture of exponentials/geometrics	T	4.11	UN 4.1(d)
sign rule for symmetric pairs; CC: $r_0 > 0 > r_1$	T	4.4	SP 5.6, 5.7
monotone signs across a pure-parity eigenvalue	T †	4.5	SP 5.9
CC residues alternate iff $d = 1$ or $N = 2$	T †	4.6	SP 5.10
sign pattern as the sign of one polynomial	T †	4.7	SP 5.11
certified sign patterns, 20 pairs (d, N)	CA	4.8	SP 14.1
“CC residues alternate in $d \geq 2$ ”	F	4.9	SP 5.10, 14.1
laziness monotonicity; unimodal set $= (0, q^*]$	T	4.10	UN 2.5, 2.6; 1D 3.8
tail (cone) lemma	T	4.12	CE 5; UN 6.1
last-exit decomposition; log-concave criterion	T	4.13, 4.14	UN 4.7, 4.8, 4.11
$\tau \leq 1/\sigma_0 \leq \tau + 1/\Lambda_1$ and refinements; bounds on σ_1, r_0, r_1	T	4.15	SP 6.2–6.8
tail of the neglected poles	T	4.16	SP 12.1, 12.2
<i>One dimension</i>			
exact law, product formula, moments	T	5.1	1D 2.2, 2.3, 2.5
continuous-time unimodality, every start (non-degenerate mode)	T	5.2	1D 3.3; UN 4.3
discrete unimodality for $q \leq q_N^ $, every start (strict; case $q = q_N^ $)	T (†)	5.3	1D 3.5; UN 4.3
log-concavity for $q \leq \frac{2}{3}$; continuous time every q	T	5.4	1D 3.6 (CE 28 for $q \leq \frac{1}{2}$)
$N = 3$: log-concave iff $q \leq 3 - \sqrt{5}$	T †	5.5	CE Rem 29(a)
not unimodal for $q > \frac{2N-4}{2N-3}$; $N = 4$ iff $q \leq \frac{4}{5}$	T	5.6	1D 3.9; UN 4.4; CE 32
$q = 1$: parity classes unimodal	T	5.7	1D 3.10
eventual TP_2 at $q = \frac{4}{5}, \frac{17}{20}, \frac{9}{10}$; unimodal for $q \leq \frac{4}{5}$, all N	CA	5.8	1D 3.12–3.16
eventual TP_2 for every $q < 1$	C	5.9	1D Rem 3.17
exact thresholds end-to-end ($N \leq 300$) and centre-to-corner ($N \leq 301$)	CA †	5.10	UN 7.3, 7.4
“ $q^*(N) = \frac{2N-4}{2N-3}$ for all $N \geq 20$ ”	F	5.11	UN 7.4
starts $x_0 \geq 2$: dip for $q > q_{\text{fail}}(N - x_0)$	T †	5.12	1D 8.8; UN 7.3
q_{fail} not sharp; exact scan at $q = \frac{4}{5}$, $N \leq 12$	CA †	5.13	1D Obs 8.9; UN Rem after 7.3

statement	status	here	source
exact scan $N \leq 13$, six values of q (no failure found in 225 cases)	O	5.14	1D Obs 8.9
limit density, unique non-degenerate maximiser c	T	5.15	1D 4.2, 4.3, 8.3
70-digit enclosure of c ; κ, ρ	CA	5.16	1D 4.6; CE 23
continuous-time mode, $ u_N - c + \kappa L^{-2} < 1.78L^{-4}$, κ_2	CA († for κ_2)	5.17	1D 5.7, 5.8
discrete-time mode window for $q \leq \frac{4}{5}$ ($N \geq 3$), $\frac{17}{20}$ ($N \geq 7$), $\frac{9}{10}$ ($N \geq 10$)	CA	5.18	1D 6.1–6.4
$q = 1$: parity of the mode, rate N^{-1}	CA	5.19	1D 7.4
global maximisers for $q < 1$, $N \geq 601$; limit theorem for every q	CA	5.20	1D 7.7, 7.8
“closed form valid in $d = 1$ ”	F (CA)	5.21	1D Rem 7.9; CE 21, 43
general start: limit density, limit theorem	T	5.22	1D 8.3, 8.5
general start: explicit law for $\xi_N \leq \frac{1}{2}$; values of $c(\xi)$	CA	5.23	1D 8.6, 8.12, 8.14
exact modes, $q = \frac{4}{5}$ ($N \leq 1500, 2000$), $\frac{1}{2}$ ($N \leq 1000$), $q = 1$ ($N \leq 400$)	CA	5.24	CE 12, 13, 30, 39, 40, 45; 1D 3.18
<i>Unimodality in every dimension</i>			
CC, C2M, M2C: continuous time, all N, d ; discrete for $q \leq q_N^\square$	T	6.1	UN 4.12 (also AS 1.11, 7.4)
location of the continuous-time mode	T	6.2	UN 4.13
1D free increment log-concave ($q \leq \frac{4}{5}$, all $N \geq 9$; iff $q \leq q_1(N)$, $N \leq 501$)	CA	6.3	UN 9.6–9.11
the value: unimodal for $q \leq \frac{4}{5}$, all N , all d (CC), $d \geq 2$ (centre)	CA	6.4	UN 9.12, 9.13
finite ranges beyond $\frac{4}{5}$; $d = 2$, $q = 1$ exceptions	CA	6.5	UN 7.1
$q = 1$, extended ranges ($d = 2$: 16 non-unimodal $N \leq 200$)	CA	6.6	CE 13, 40
“always unimodal”	F	6.7	UN 8; CE 9
explicit tail and rise windows, CC, every q	T	6.8	UN 5.6, 5.10
explicit bimodal pairs ($d = 3$, $N = 3$; $d = 2$, $N = 8$)	CA	6.9	UN 8.1
exhaustive classification of small boxes	CA	6.10	UN 8.2
M2C unimodal but not log-concave	T	6.11	UN 8.4
log-concavity/front property; ≤ 3 sign changes; peak-last	C	6.12	UN 10.1, 10.3; CE 47
exact modes $d = 2, 3$, $q \in \{\frac{4}{5}, \frac{1}{2}\}$; q -scaling	CA	6.13	CE 12, 18, 39, 42
<i>Spectral quantities, CC</i>			
arrival profile; $\frac{d(N^2-1)}{3} \leq m - \tau \leq \frac{1+\ln W}{\Lambda_1}$	T	7.1	SP 7.2
second-moment functional; size of $\bar{X}$	T	7.2	SP 7.4, 7.7
rational enclosure of the MFPT	T	7.3	CE 19
Epstein-type sums Z_d, Z'_d	CA	7.4	SP 7.6
monotone coupling	T	7.5	SP 7.8; UN 5.1
$r_0 > 1$, $\sigma_0 m > 1$	T	7.6	SP 7.9
explicit bounds in $\bar{X}$, every $d \geq 2$	T	7.7	SP 7.10, 7.11
single and double sums for τ, m	T	7.8	SP 8.1–8.5
$d = 2$: $m_2 = \frac{8}{\pi} N^2 \ln N + C_2 N^2 + O(1)$, $ R_m \leq 11.3$	CA	7.9	SP 9.1
$d = 2$: constant terms, $C_2^{(0)} = 0.5321408 \dots$	T † (values CA)	7.10	SP 9.6, 9.8
closed form of C_2	CA	7.11	CE 25
C_2 is the N^2 coefficient (two closed forms agree)	T ‡ (explicit remainder CA)	7.12	SP 9.1, 9.2; CE 25
$0.3211 < r_N < 0.53213$, increasing ($N \leq 301$); up to $N = 1500$	CA ($^\ddagger, ^\dagger$)	7.13	CE 26, 27; SP 8.3
$d = 3$: C_3 , second order, remainders, constant terms	CA (limits T)	7.14	SP 10.1
$\bar{X}, \Lambda_1(m - \tau)$ explicit in N	CA	7.15	SP 11.1, 11.2
truncated-constant forms	F (CA) †	7.16	SP 11.5
first-order pole structure	T	7.17	SP 11.3
<i>The mode in $d = 2, 3$ and the closed form</i>			
decomposition $\mu T = U + \mu T_\pi$; pole expansion	T	8.1	AS 1.2–1.7
continuous-time unimodality, g'/φ nonincreasing	T († extension)	8.2	AS 1.10, 1.11

statement	status	here	source
exact identity and window theorem	T	8.3	AS 2.1–2.3
pole data in terms of X_s	T	8.4	AS 3.1–3.5
lattice sums, bounds on X_s , X , $\mathbb{E}U$	CA	8.5	AS 4.5–4.7
closed form, continuous time: $ X(\tau^* - \tau_{\text{cf}}) $ bounded, all $N \geq N_0$	CA	8.7	AS 5.1
asymptotic laws with explicit remainders, $N \geq 20$	CA	8.8	AS 6.1–6.4
discrete time, $q \leq \frac{1}{2}$ (unimodality input by hand)	CA [†]	8.9	AS 7.7–7.9; UN 4.12
localisation without unimodality	T	8.10	AS 7.10–7.13
discrete time, $q \leq q_{\max} \in \{0.8, 0.9, 0.95, 0.99\}$	CA	8.11	AS 7.14, 7.15
constants uniform in q	T [†]	8.12	AS 7.16
discrete time, every $q < 1$, $N \geq \max(N_*, N_1(q))$	CA [†]	8.13	AS 7.17, 7.18
closed form within 0.970% ($d = 2$, $q = \frac{4}{5}$; 0.971% for $q = \frac{1}{2}$), 0.275% ($d = 3$) on finite ranges	CA	8.14	CE 21, 43
closed form within 0.3% for all $N \geq 10$ ($d = 3$); 1% for large N ($d = 2$)	CA [†]	8.15	this document
$q = 1$ laws; 1% for all $N \geq 10$ in $d = 2$	C	8.16	AS 7.19; CE 48
limits $K_2^\infty \approx 3.74$, $K_3^\infty \approx -0.22$	O	8.17	AS 5.3
<i>Open problems</i>			
unimodality thresholds for $\frac{4}{5} < q < 1$	C	12.1	UN 10.2, 10.4
further numerical observations: thresholds by bisection (1D 3.20, UN 4.5), $d = 2$ remainders monotone (SP 9.7), $d = 3$ remainders (SP 10.12), residue statistics (SP 14.2), log-concavity in $d = 2, 3$ (CE 49), front property (UN 6.2)	O	—	as listed

4 Results for every finite symmetric chain and every start–target pair

Throughout this section Ω is a finite set with $n \geq 2$ elements and $\mathcal{L}$ a real symmetric matrix with $\mathcal{L}_{xy} \leq 0$ for $x \neq y$, $\mathcal{L}\mathbf{1} = 0$, and connected graph $\{x \sim y \iff \mathcal{L}_{xy} < 0\}$. The box generator of Definition 2.1 is such a matrix. Activity q enters only through the scalings $q\sigma$ (continuous time) and $1 - q\sigma$ (discrete time); the discrete walk is $P_q = I - q\mathcal{L}$, stochastic when $q \max_x \mathcal{L}_{xx} \leq 1$. The distinct eigenvalues of $\mathcal{L}$ are $0 = \lambda^{(0)} < \dots < \lambda^{(M)}$ with spectral projectors E_i . An eigenvalue is *visible from a* if $w_i := \|E_i e_a\|^2 > 0$; the visible ones are listed as $0 = \Lambda_0 < \Lambda_1 < \dots < \Lambda_p$. The secular and cross functions are

$$G(s) = \langle e_a, (\mathcal{L} - s)^{-1} e_a \rangle = \sum_{i=0}^p \frac{w_i}{\Lambda_i - s}, \quad G_x(s) = \langle e_x, (\mathcal{L} - s)^{-1} e_a \rangle.$$

4.1 Spectral representation, interlacing, residues

Theorem 4.1 (structure and first-passage representation; [SP, Thms 2.2, 3.1]). *(i) G has exactly p zeros $\sigma_0 < \dots < \sigma_{p-1}$; they are real, simple and strictly interlace the visible eigenvalues, $0 = \Lambda_0 < \sigma_0 < \Lambda_1 < \sigma_1 < \dots < \sigma_{p-1} < \Lambda_p$. (ii) The σ_j are the eigenvalues of $\mathcal{L}_a$ on the cyclic subspace of e_a intersected with $e_a^\perp$; all other eigenvalues of $\mathcal{L}_a$ are eigenvalues of $\mathcal{L}$ with eigenfunctions vanishing at a , and $\det(\mathcal{L}_a - s) = G(s) \det(\mathcal{L} - s)$. (iii) With the residues $r_j(x) := -G_x(\sigma_j)/(\sigma_j G'(\sigma_j))$,*

$$\mathbb{P}_x(T > t) = \sum_{j=0}^{p-1} r_j(x) e^{-q\sigma_j t}, \quad f(t) = \sum_j r_j(x) (1 - \nu_j) \nu_j^{t-1} \quad (\nu_j := 1 - q\sigma_j, \quad 0^0 := 1),$$

in continuous and in discrete time respectively, and $\sum_j r_j \sigma_j / (\sigma_j - s) = G_x(s)/G(s)$ (the renewal ratio of the article). The pole σ_j is absent from the law started at x iff $G_x(\sigma_j) = 0$.

Proof (outline). On each interval $(\Lambda_j, \Lambda_{j+1})$ the function G increases strictly from $-\infty$ to $+\infty$, and $G \prod_l (\Lambda_l - s)$ is a polynomial of degree p ; this gives (i). The vectors $\psi_j = \sum_i (\Lambda_i - \sigma_j)^{-1} E_{\Lambda_i} e_a$ satisfy $(\mathcal{L} - \sigma_j) \psi_j = e_a$, $\psi_j(a) = G(\sigma_j) = 0$, hence $\mathcal{L}_a \psi_j = \sigma_j \psi_j$, and they form an orthogonal basis of the invariant subspace that contains $\mathbf{1}_{\Omega'}$; expanding $\mathbf{1}_{\Omega'}$ in this basis gives (iii). Complete proof: [SP, Sections 2–3]. $\square$

Theorem 4.2 (interlacing; [SP, Thm 4.1, Cor 4.2]). *The eigenvalues $\mu_1 \leq \dots \leq \mu_{n-1}$ of $\mathcal{L}_a$ interlace those of $\mathcal{L}$ (Cauchy; R. A. Horn, C. R. Johnson, Matrix Analysis, 2nd ed., Thm 4.3.17). For the box and a corner*

target every eigenvalue of $\mathcal{L}$ is visible, the number of poles is the number of distinct values of λ_k minus one, all eigenfunctions ψ_j are symmetric under coordinate permutations, and for $d \geq 2$

$$\Lambda_1 = \frac{1}{d}(1 - \cos \frac{\pi}{N}) \text{ (multiplicity } d), \quad \Lambda_2 = 2\Lambda_1, \quad 0 < \sigma_0 < \Lambda_1 < \sigma_1 < 2\Lambda_1 < \sigma_j \text{ (} j \geq 2).$$

Theorem 4.3 (residues; [SP, Thm 5.1, Props 5.4, 5.5], [UN, Lemma 3.1, Thm 3.2]). *Let $x \neq a$ and $r_j = r_j(x)$. (a) $\sum_j r_j = 1$, and $\mathbb{E}_x T = q^{-1} \sum_j r_j / \sigma_j$. (b) If the graph of $\mathcal{L}_a$ is connected, σ_0 is the smallest eigenvalue of $\mathcal{L}_a$, it is simple, its eigenvector has a strict sign, and $r_0(y) > 0$ for every $y \in \Omega'$. (c) With $D = \text{dist}(x, a)$: $\sum_j r_j \sigma_j^i = 0$ for $1 \leq i \leq D-1$ and $(-1)^{D-1} \sum_j r_j \sigma_j^D > 0$. Consequently the residue sequence has at least $D-1$ sign changes. For the segment from the reflecting end, $p = D = N-1$ and $\text{sign } r_j = (-1)^j$. (d) From the uniform start the law is a mixture of exponentials (geometrics) with positive weights $\varrho_j = 1/(n\sigma_j^2 G'(\sigma_j))$, $\sum_j \varrho_j = 1 - \frac{1}{n}$; for every start $\sum_j |r_j(x)| \leq \sqrt{n-1}$. (e) From the quasi-stationary distribution $\nu \propto \psi_0$ the hitting time is exactly exponential (geometric): $\sum_x \nu(x) r_j(x) = \delta_{j0}$ and $\sum_x \nu(x) m(x) = 1/\sigma_0$.*

Proof (outline). (a) is (iii) of Theorem 4.1 at $t = 0$ and integration. (b) is the Perron–Frobenius theorem for $cI - \mathcal{L}_a$ (Horn–Johnson, Thm 8.4.4; the case $|\Omega'| = 1$ is trivial and treated separately in the proof). (c) $\sum_j r_j \sigma_j^i = (\mathcal{L}_a^i \mathbf{1}_{\Omega'})(x)$ is a sum over paths of length $i-1$ from x to a neighbour of a , which vanishes for $i < D$ and has the sign $(-1)^{D-1}$ for $i = D$; a polynomial of degree $\leq D-1$ vanishing at 0 and separating the sign blocks then gives the sign changes. (d), (e): orthogonality of the ψ_j and Cauchy–Schwarz. Complete proofs: [SP, Section 5]; (c) also [UN, Thm 3.2]. $\square$

4.2 Reflection-symmetric pairs: the signs of the residues

Theorem 4.4 (sign rule; [SP, Thm 5.6, Cor 5.7]). *Let R be an involutive permutation of Ω commuting with $\mathcal{L}$ with $Ra = x_0 \neq a$, $v_{\pm} = \frac{1}{2}(e_a \pm e_{x_0})$ and $G^{\pm}(s) = \langle v_{\pm}, (\mathcal{L} - s)^{-1} v_{\pm} \rangle$. Then $G = G^+ + G^-$ and*

$$r_j(x_0) = \frac{2G^-(\sigma_j)}{\sigma_j G'(\sigma_j)} = -\frac{2G^+(\sigma_j)}{\sigma_j G'(\sigma_j)}.$$

If moreover the first visible eigenvalue Λ_1 is purely odd ($E_{\Lambda_1} v_+ = 0$) and $\mathcal{L}$ has at least two eigenvalues (with multiplicity) in $(0, \Lambda_1]$, then $r_1(x_0) < 0$. In particular, for the box with $d \geq 2$, every $N \geq 2$ and the CC pair: $r_0 > 0 > r_1$.

The natural guess that, as for the segment (Theorem 4.3(c)), the residues of the CC pair alternate in sign in every dimension is false (Refuted statement 4.9). The following three statements of SP were proved after its internal checks; their proofs are reproduced in full.

Proposition 4.5 (monotone signs across an eigenvalue of pure parity; [SP, Prop 5.9]). † *In the situation of Theorem 4.4 call a visible eigenvalue Λ_i purely even if $E_{\Lambda_i} v_- = 0$ and purely odd if $E_{\Lambda_i} v_+ = 0$. Let $1 \leq i \leq p-1$ and $r_j = r_j(x_0)$. (a) If Λ_i is purely even, then $G^-(\sigma_{i-1}) < G^-(\sigma_i)$; hence (r_{i-1}, r_i) does not have the signs $(+, -)$. (b) If Λ_i is purely odd, then $G^+(\sigma_{i-1}) < G^+(\sigma_i)$; hence (r_{i-1}, r_i) does not have the signs $(-, +)$. Over a run of consecutive purely even (purely odd) eigenvalues the signs of the adjacent residues are non-decreasing (non-increasing).*

Proof. R commutes with every spectral projector and $Rv_{\pm} = \pm v_{\pm}$, so $\langle E_i v_+, E_i v_- \rangle = 0$ and $\|E_i e_a\|^2 = \|E_i v_+\|^2 + \|E_i v_-\|^2$; every pole of $G^{\pm}$ is therefore a visible eigenvalue. (a) If $E_{\Lambda_i} v_- = 0$, the only visible eigenvalue in $I = (\Lambda_{i-1}, \Lambda_{i+1})$ is not a pole of G^- , so G^- is real-analytic on I with derivative $\sum_l \|E_l v_-\|^2 (\lambda^{(l)} - s)^{-2} > 0$ (some weight is positive since $\sum_l \|E_l v_-\|^2 = \|v_-\|^2 = \frac{1}{2}$). By Theorem 4.1(i), $\sigma_{i-1} < \Lambda_i < \sigma_i$ both lie in I , so $G^-(\sigma_{i-1}) < G^-(\sigma_i)$; and $\text{sign } r_j = \text{sign } G^-(\sigma_j)$ because $\sigma_j G'(\sigma_j) > 0$. If $r_{i-1} > 0$ then $r_i > 0$. (b) is the same argument for G^+ with $\text{sign } r_j = -\text{sign } G^+(\sigma_j)$ ($0 = \Lambda_0 \notin I$). The last sentence follows by applying (a) or (b) to consecutive indices. $\square$

Theorem 4.6 (the CC residues alternate iff $d = 1$ or $N = 2$; [SP, Thm 5.10]). † *For the box with $a = (N, \dots, N)$, $x_0 = (1, \dots, 1)$, $D = d(N-1)$: (a) if $d = 1$ or $N = 2$, then $p = D$ and $\text{sign } r_j = (-1)^j$ for all j ; (b) if $d \geq 2$ and $N \geq 3$, put $i = p-3$ ($N \neq 4$), $i = p-4$ ($N = 4$). Then $i \geq 1$, and Λ_i, Λ_{i+1} are both purely even or both purely odd; the triple $(\text{sign } r_{i-1}, \text{sign } r_i, \text{sign } r_{i+1})$ is monotone, so the residues do not alternate; (c) moreover (r_2, r_3) is not $(+, -)$ for $d = 2$, $N \geq 3$; (r_4, r_5) is not $(+, -)$ for $d \geq 4$, $N \geq 4$; (r_6, r_7) is not $(+, -)$ for $d = 3$, $N \geq 7$. The proof uses no computation.*

Proof. Let $u = \frac{\pi}{2N}$, $x = \sin^2 u$, $\varsigma_j = \sin^2(ju) / \sin^2 u$ ($0 \leq j \leq N-1$), strictly increasing in j , with $\varsigma_1 = 1$, $\varsigma_2 = 4(1-x) =: t$, $\varsigma_3 = (3-4x)^2$. Then $\lambda_k = \Lambda_1 V(k)$, $V(k) = \sum_i \varsigma_{k_i}$, and since every eigenvalue is visible from the corner, $\Lambda_0 < \dots < \Lambda_p$ is the list of distinct values of V times Λ_1 . With $R : x_i \mapsto N+1-x_i$,

$\phi_k \circ R = (-1)^{|k|} \phi_k$; if all k with $\lambda_k = \Lambda_i$ have $|k|$ of one parity, Λ_i is purely even (even parity) or purely odd (odd parity).

(a) $d = 1$ is Theorem 4.3(c). For $N = 2$, $V(k) = |k| \in \{0, \dots, d\}$, so $p = d = D$; at least $D - 1 = p - 1$ sign changes force alternation, and $r_0 > 0$.

(b) Count from the top. Put $\Delta_m = \varsigma_{N-1} - \varsigma_{N-1-m}$ ($0 \leq m \leq N - 1$). Since $\sin^2 A - \sin^2 B = \sin(A + B) \sin(A - B)$ and $(2N - 2m - 1)u = \pi - (2m + 1)u$, $\Delta_m - \Delta_{m-1} = \sin((2m + 1)u) / \sin u > 0$, so Δ_m is strictly increasing. For $m_l := N - 1 - k_l$ and $\Delta(m) = \sum_l \Delta_{m_l}$ we have $V(k) = d\varsigma_{N-1} - \Delta(m)$ and $|k| = D - |m|$; hence $\Lambda_{p-j} = \Lambda_1(d\varsigma_{N-1} - \Delta^{(j)})$ for the distinct values $0 = \Delta^{(0)} < \Delta^{(1)} < \dots$ of Δ , and Λ_{p-j} is pure if all m with $\Delta(m) = \Delta^{(j)}$ have $|m|$ of one parity. Let $\alpha = \Delta_1 = \sin 3u / \sin u = 3 - 4x$ and $\beta = \Delta_2 - \Delta_1 = \sin 5u / \sin u = 5 - 20x + 16x^2$. For $N \geq 3$, $0 < x \leq \frac{1}{4}$, so $2\alpha - \beta = 1 + 12x - 16x^2 > 0$, and $\beta - \alpha = 2(1 - 8x + 8x^2)$ is positive for $N \geq 5$ and zero for $N = 4$. $N \geq 5$: $0 < \alpha < \beta < 2\alpha$. If $\Delta(m) \leq \alpha + \beta = \Delta_2$, all entries of m are ≤ 2 ; an entry 2 forces $m = \{2\}$; otherwise $\Delta(m) = n_1\alpha$ with $n_1 \leq 2$. So the four smallest values are $0 < \alpha < 2\alpha < \alpha + \beta$, attained only by $\emptyset, \{1\}, \{1, 1\}, \{2\}$; Λ_{p-2} and Λ_{p-3} are pure with $|m| = 2$. $N = 3$: $\Delta_1 = 2$, $\Delta_2 = 3$, $\Delta(m) = 2n_1 + 3n_2$, four smallest values $0 < 2 < 3 < 4$ from $\emptyset, \{1\}, \{2\}, \{1, 1\}$; $\Lambda_{p-2}, \Lambda_{p-3}$ pure with $|m| = 2$. $N = 4$: $x = \sin^2 \frac{\pi}{8}$, $\alpha = \beta = 1 + \sqrt{2}$, $\Delta_3 - \Delta_2 = \sin \frac{7\pi}{8} / \sin \frac{\pi}{8} = 1$, so $\Delta(m) = c\alpha + n_3$ with $c = n_1 + 2n_2 + 2n_3$ and $|m| = c + n_3$. As α is irrational, $\Delta(m)$ determines (c, n_3) , hence $|m|$: every eigenvalue is pure. The values below 3α are $0 < \alpha < 2\alpha < 2\alpha + 1$ ($2n_3 \leq c \leq 2$, $\alpha > 1$), and 3α is attained ($\{1, 2\}$, $d \geq 2$). So Λ_{p-3} ($(c, n_3) = (2, 1)$) and Λ_{p-4} ($(3, 0)$) are pure with $|m| = 3$. *Conclusion*: Λ_0 corresponds to $m = (N - 1, \dots, N - 1)$ with $\Delta(m) = d\Delta_{N-1} \geq 2\Delta_2$, larger than all values found; hence $p \geq 4$ ($N \neq 4$), $p \geq 5$ ($N = 4$), so $i \geq 1$ and $i + 1 \leq p - 2$. Λ_i, Λ_{i+1} are pure of the same parity (that of $D - 2$, resp. $D - 3$), and Proposition 4.5 with two consecutive pure eigenvalues makes the triple monotone, whereas alternation needs $(+, -, +)$ or $(-, +, -)$.

(c) $d = 2$, $N \geq 3$: $t \geq 3$, values of V with all components ≤ 2 are $0, 1, 2, t, t + 1, 2t$, so $\Lambda_3 = t\Lambda_1$, spanned by $\phi_{(2,0)}, \phi_{(0,2)}$ ($|k| = 2$): purely even, and $p - 1 \geq 3$. By Proposition 4.5(a), $r_2 \geq 0 \Rightarrow r_3 > 0$. $d \geq 4$, $N \geq 4$: $3 < t < 4$, $\varsigma_3 > 5.8$; the six smallest values are $0 < 1 < 2 < 3 < t < 4$ and $\Lambda_5 = 4\Lambda_1$ is spanned by the ϕ_k , k a permutation of $(1, 1, 1, 1, 0, \dots)$: purely even. $d = 3$, $N \geq 7$: $3.8 < t < 4$, $\varsigma_3 > 2t$; the eight smallest values are $0 < 1 < 2 < 3 < t < t + 1 < t + 2 < 2t$, $\Lambda_7 = 2t\Lambda_1$ spanned by permutations of $(2, 2, 0)$: purely even. Apply Proposition 4.5(a) in each case. $\square$

Remark 4.7 (the sign pattern as the sign of one polynomial; [SP, Rem 5.11]). † For any chain and $x \neq a$ with $D = \text{dist}(x, a)$, $\omega_i = \langle e_x, E_{\Lambda_i} e_a \rangle$, the polynomial $\mathcal{N}_x(s) = \sum_i \omega_i \prod_{l \neq i} (\Lambda_l - s)$ has degree exactly $p - D$ and leading coefficient $(-1)^p (\mathcal{L}^D)_{xa}$ (of sign $(-1)^{p+D}$), and $\text{sign } r_j(x) = (-1)^j \text{sign } \mathcal{N}_x(\sigma_j)$. Indeed $G_x(s) = -\sum_{m \geq D} (\mathcal{L}^m)_{xa} s^{-m-1}$ for large $|s|$ and $\prod_l (\Lambda_l - s) = (-1)^{p+1} s^{p+1} (1 + O(s^{-1}))$; and exactly $j + 1$ factors of $\prod_l (\Lambda_l - \sigma_j)$ are negative. In particular the residues alternate whenever $p = D$, and at most $p - D$ pairs (r_j, r_{j+1}) of non-zero residues fail to alternate (such a pair needs a zero of $\mathcal{N}_x$ in (σ_j, σ_{j+1})).

Computer-assisted proposition 4.8 (certified sign patterns; [SP, Prop 14.1]). *For the CC pair the signs of all residues are certified (exact grouping of the eigenvalues modulo the cyclotomic polynomial Φ_{2N} , then ball arithmetic) for the 20 pairs $(d, N) \in \{2\} \times \{2, \dots, 8, 10, 11, 12\} \cup \{3\} \times \{2, \dots, 6\} \cup \{4\} \times \{2, 3, 4\} \cup \{5\} \times \{2, 3\}$; e.g. $(2, 3)$: $+-+-$, $(2, 4)$: $+++++-$, $(3, 3)$: $+-+-$. All residues are non-zero there, the sequence alternates for $N = 2$ and not for $N \geq 3$, consistently with Theorem 4.6.*

Refuted statement 4.9 ([SP, Section 5.2, Prop 14.1]). “The residues of the corner-to-corner law alternate in sign in $d \geq 2$, as they do in $d = 1$.” False for every $d \geq 2$, $N \geq 3$ (Theorem 4.6(b)); the smallest example is $d = 2$, $N = 3$, signs $+-+-$ (Computer-assisted proposition 4.8).

Descartes’ rule applied to the residues therefore cannot prove unimodality in $d \geq 2$: it bounds the number of zeros of g'_q by the number of sign changes of the residues, which is at least $D - 1$ [UN, Section 3]. The unimodality results below use different tools.

4.3 Laziness monotonicity

For a fixed pair let $a_q(t) := f_q(t + 1)$ and $\Delta a_q(t) = a_q(t + 1) - a_q(t)$; $S^-(\cdot)$ is the number of sign changes. With $p = q'/q$ and the binomial weights $b_p(t, n) = \binom{t}{n} p^n (1 - p)^{t-n}$, $Q_{q'} = (1 - p)I + pQ_q$ gives

$$\Delta a_{q'}(t) = p^2 \sum_{n \leq t} b_p(t, n) \Delta a_q(n), \quad g'_q(s) = (q/q_0)^2 e^{-\mu} \sum_{n \geq 0} \frac{\mu^n}{n!} \Delta a_{q_0}(n) \quad (\mu = qs/q_0). \quad (1)$$

Theorem 4.10 (laziness monotonicity; [UN, Thm 2.5, Cor 2.6]; for the segment [1D, Lemma 3.8]). *For a fixed box, target and start: $S^-(\Delta a_{q'}) \leq S^-(\Delta a_q)$ for $0 < q' \leq q \leq 1$, and g'_q has at most $S^-(\Delta a_{q_0})$ zeros in $(0, \infty)$ for every rate q and every $q_0 \in (0, 1]$. Consequently the set of $q \in (0, 1]$ for which f_q is unimodal is empty or an interval $(0, q^*]$; if it is non-empty the continuous-time density is unimodal; and the number of local maxima of f_q is non-increasing as q decreases.*

Proof (outline). The binomial matrix $(b_p(t, n))$ is a limit of products of nonnegative bidiagonal matrices (Pascal matrix = $\exp(H)$ with H nilpotent bidiagonal), and such matrices do not increase the number of sign changes (an interleaving argument, self-contained); the Poisson form (1) and the Descartes–Laguerre rule for power series give the continuous-time statement. Complete proof: [UN, Section 2]. For the segment from the reflecting end a probabilistic proof (thinning by an independent Bernoulli sequence) is [1D, Lemma 3.8]. $\square$

Theorem 4.11 (a single distant start is never a mixture; [UN, Thm 4.1(d)]). *For the box, any target a and a single start at distance $D \geq 2$ there is no positive Borel measure μ on $[0, 1)$ with $f_q(t) = \int \lambda^{t-1} \mu(d\lambda)$ for all $t \geq 1$, and no positive measure on $[0, \infty)$ with $g_q(t) = \int e^{-\sigma t} \mu(d\sigma)$ for $t > 0$. In particular some residue is negative.*

Proof. $D \geq 2$ gives $f_q(1) = q r(x_0) = 0$ and $g_q(0) = 0$, while $f_q(D) \geq (q/2d)^D > 0$ (a shortest path) and $g_q(t) > 0$ for $t > 0$ (by (1) above g_q is a Poisson mixture of the values $f_{q_0}(n+1) \geq 0$, one of which is positive). A nonnegative mixture $\int \lambda^{t-1} \mu(d\lambda)$ equals $\mu([0, 1))$ at $t = 1$, so $\mu = 0$ and $f_q \equiv 0$, a contradiction; in continuous time $\mu([0, \infty)) = \lim_{t \downarrow 0} g_q(t) = 0$ by monotone convergence. By Theorem 4.3(c) the residues have at least $D - 1 \geq 1$ sign changes, so some residue is negative. $\square$

4.4 The tail lemma (cone certificate)

Lemma 4.12 (tail lemma; [CE, Lemma 5], [UN, Lemma 6.1]). *Let $v_t = Q_q^{t-1} e_{x_0}$ (occupation probabilities of the surviving walker, a symmetric Q_q). If $v_{T+1} < v_T$ at every site of Λ' for some $T \geq 1$, then $v_{t+1} < v_t$ for all $t \geq T$ and $f_q(t+1) < f_q(t)$ for all $t \geq T$. If moreover $v_{T+1} \leq \gamma v_T$ with $\gamma < 1$, then $f_q(T+s) \leq \gamma^s f_q(T)$. (With $\leq$ in place of $<$ the conclusions hold with $\leq$.)*

Proof. $w_t := v_t - v_{t+1}$ satisfies $w_{t+1} = Q_q w_t$; every row of Q_q has a positive entry, so $w_T > 0$ propagates to $w_t > 0$ for $t \geq T$, and $f_q(t) - f_q(t+1) = \frac{q}{2d} \sum_{y \sim a} w_t(y) > 0$. For the geometric bound, $v_{T+s+1} = Q_q^s v_{T+1} \leq \gamma Q_q^s v_T = \gamma v_{T+s}$ since $Q_q \geq 0$. $\square$

A tail time always exists [CE, Prop 6]: $v_{t+1}(y)/v_t(y) \rightarrow \lambda_1(Q_q) < 1$ for every y (Perron–Frobenius for the primitive symmetric matrix Q_q , proved in a few lines there; also [UN, Lemma 3.1]). Every PMF is therefore eventually strictly decreasing, and a failure of unimodality can only occur in a bounded window.

4.5 Last-exit decomposition and a criterion for unimodality

Let X be the unkilld walk, $u(t) = \mathbb{P}_{x_0}(X_t = a)$ the free propagator, and in discrete time $\varphi(t) = u(t) - u(t-1)$ ($u(-1) := 0$), $E(k) = \mathbb{P}_a(X_1 \neq a, \dots, X_k \neq a)$, $\nu(k) = E(k) - E(k+1)$.

Lemma 4.13 (last-exit decomposition; [UN, Lemma 4.7]). *For every finite Markov chain and every pair: in discrete time $f_q(t) = \sum_{k=0}^t E(k) \varphi(t-k)$ and $f_q(t+1) - f_q(t) = \varphi(t+1) - \sum_{k \leq t} \nu(k) \varphi(t-k)$; in continuous time $g_q = u' + r * u'$ with $r(\tau) = q r^\top e^{-q\tau} \mathbf{1}$ positive and nonincreasing.*

Theorem 4.14 (criterion; [UN, Lemma 4.8, Thm 4.11]). *(a) If $u' > 0$ on $(0, \infty)$ and $\log u'$ is concave there, then the continuous-time density g_q is unimodal for every rate, with a unique mode; more precisely $g'_q = u' \cdot B$ with B nonincreasing. (b) If for some q the sequence φ is log-concave, then f_q is unimodal.*

Proof (outline). A log-concave function convolved with a kernel consisting of an atom at 0 plus a nonincreasing density is unimodal: in $G = \varphi + r * \varphi$ one has $G' = \varphi B$ with $B = \varphi'/\varphi + r(0) - \int_0^t \nu(\tau) \varphi(t-\tau)/\varphi(t) d\tau$, and each ratio $\varphi(t-\tau)/\varphi(t)$ is nondecreasing in t by log-concavity (the elementary half of Ibragimov’s strong-unimodality theorem; the discrete case likewise). Complete proof: [UN, Section 4.4]. $\square$

4.6 Explicit bounds on the principal pole and the neglected poles

With $\tau = n \sum_{i \geq 1} w_i / \Lambda_i$ (the mean hitting time from the uniform start, [SP, Lemma 6.1]), $K_2 = n \sum_{i \geq 1} w_i / \Lambda_i^2$ and $\varkappa = K_2 / \tau^2 \leq 1 / \bar{X}$:

Theorem 4.15 (principal killed eigenvalue; [SP, Thm 6.2]). *$\tau \leq 1/\sigma_0 \leq \tau + 1/\Lambda_1$, i.e. $\bar{X}/(1+\bar{X}) \leq \sigma_0 \tau \leq 1$. If $\bar{X} > 1$ and $\beta = \bar{X}/(\bar{X}-1)$, then $1 - \beta \varkappa \leq 2/(1 + \sqrt{1 + 4\beta \varkappa}) \leq \sigma_0 \tau \leq 2/(1 + \sqrt{1 + 4\varkappa}) \leq 1 - \varkappa + 2\varkappa^2$.*

Proof. σ_0 is the zero of G in $(0, \Lambda_1)$, i.e. $1 = \sigma_0 n g(\sigma_0)$ with $g(s) = \sum_{i \geq 1} w_i / (\Lambda_i - s)$. For $0 \leq s < \Lambda_1$, $\tau + sK_2 \leq n g(s) \leq \tau + sK_2/(1 - s/\Lambda_1)$ and $n g(s) \leq \tau/(1 - s/\Lambda_1)$, from $\frac{1}{\Lambda} + \frac{s}{\Lambda^2} \leq \frac{1}{\Lambda-s} \leq \frac{1}{\Lambda} + \frac{s}{\Lambda^2(1-s/\Lambda_1)}$ for $\Lambda \geq \Lambda_1$. At $s = \sigma_0$, with $t = \sigma_0 \tau$: the first inequality gives $1 \geq t + \varkappa t^2$, the third $1 - \sigma_0/\Lambda_1 \leq t$, and if $\bar{X} > 1$ the second gives $1 \leq t + \beta \varkappa t^2$. The outer inequalities are elementary. $\square$

Explicit two-sided bounds on the second pole σ_1 , the gap $\sigma_1 - \sigma_0$ and the residues r_0, r_1 in terms of three Green's-function functionals are [SP, Thms 6.5, 6.7, 6.8, Cor 6.6]; they are specialised to the box in Theorem 7.7.

Theorem 4.16 (tail of the neglected poles; [SP, Thm 12.1, Cor 12.2]). *For $1 \leq J \leq p-1$, $t_0 > 0$, unit rate: $|\mathbb{P}_x(T > t) - \sum_{j < J} r_j(x) e^{-\sigma_j t}| \leq (n p_{t_0}(x, x) \sum_{j \geq J} \varrho_j)^{1/2} e^{-\sigma_J(t-t_0/2)}$ for $t \geq t_0/2$, and the analogous bounds for the density and in discrete time hold. For the CC pair, $d \geq 2$, $\bar{X} > 1$, $c_* = 1 + \pi^{3/2}/2$, $Z \geq \Lambda_1^2 K_2$, and the continuous-time density f at unit rate: $|f(t) - r_0 \sigma_0 e^{-\sigma_0 t} - r_1 \sigma_1 e^{-\sigma_1 t}| \leq 2e c_*^{d/2} \beta \sqrt{Z} \bar{X}^{-1} \Lambda_1 e^{-2\Lambda_1 t}$ for $t \geq 1/\Lambda_1$. This bound is absolute; it is a relative error $O(\bar{X}^{-2})$ only where $e^{-\Lambda_1 t} \asymp 1/\bar{X}$, that is near the mode, whereas at $t = 1/\Lambda_1$ it is of the same order as the two slowest terms.*

5 One dimension

In this section $d = 1$: the walk lives on the segment $\{1, \dots, N\}$, site 1 is reflecting and the target is $a = N$; the start is x_0 , $d_0 = N - x_0$, and $L = N - \frac{1}{2}$. (A target in the interior reduces to this case: the walk from $x_0 < a$ never leaves $\{1, \dots, a-1\}$ before absorption.) The killed matrix $Q = Q_q$ on $\{1, \dots, N-1\}$ has $Q_{11} = 1 - \frac{q}{2}$, $Q_{jj} = 1 - q$ ($j \geq 2$), $Q_{j,j\pm 1} = \frac{q}{2}$. Put

$$\theta_m = \frac{(2m-1)\pi}{2N-1} = \frac{(2m-1)\pi}{2L}, \quad \lambda_m = 1 - q(1 - \cos \theta_m) \quad (m = 1, \dots, N-1).$$

5.1 The exact law

Theorem 5.1 (exact law; [1D, Thms 2.2, 2.3, Cor 2.5]). *For every start and $t \geq 1$, $s \geq 0$:*

$$f_{x_0}(t) = \frac{2q}{2N-1} \sum_{m=1}^{N-1} \sin \theta_m \sin(d_0 \theta_m) \lambda_m^{t-1}, \quad g_{x_0}(s) = \frac{2q}{2N-1} \sum_{m=1}^{N-1} \sin \theta_m \sin(d_0 \theta_m) e^{-q(1-\cos \theta_m)s};$$

$f_{x_0}(t) = 0$ for $t < d_0$ and $f_{x_0}(d_0) = (q/2)^{d_0}$. From the reflecting end, $\mathbb{E}z^T = \prod_{m=1}^{N-1} (1 - \lambda_m)z / (1 - \lambda_m z)$: in continuous time T is a sum of $N-1$ independent exponential variables, and for $q \leq q_N^\perp$ in discrete time a sum of $N-1$ independent geometric variables. $\mathbb{E}_{x_0}T = (N(N-1) - x_0(x_0-1))/q$; from the reflecting end $\mathbb{E}T = (L^2 - \frac{1}{4})/q$.

Proof (outline). $v_m(j) = \cos(\theta_m(j - \frac{1}{2}))$ are eigenvectors of Q ($v_m(0) = v_m(1)$ encodes the reflection, $v_m(N) = 0$ the absorption), with $\|v_m\|^2 = (2N-1)/4$; the generating function is Cramer's rule for the tridiagonal matrix $I - zQ$. These are the classical Karlin–McGregor/Keilson representations of birth–death passage times (J. A. Fill, *J. Theoret. Probab.* 22 (2009), Thms 1.1, 1.2); the proof in [1D, Section 2] is self-contained. $\square$

5.2 Unimodality and log-concavity

Theorem 5.2 (continuous time; [1D, Thm 3.3], [UN, Thm 4.3(ii)]). *For every start: if $d_0 = 1$, g_{x_0} is completely monotone; if $d_0 \geq 2$, $g_{x_0} > 0$ on $(0, \infty)$ and g'_{x_0} has exactly one zero t_c in $(0, \infty)$, with $g''_{x_0}(t_c) < 0$. More generally, for $d = 1$ and every start and target the continuous-time density is unimodal.*

Proof. By Theorem 5.1, $g = \frac{2q}{2N-1} \sum_m c_m e^{-\nu_m t}$ with $c_m = \sin \theta_m \sin(d_0 \theta_m)$ and strictly increasing ν_m . The sequence $(\sin(d_0 \theta_m))_m$ has at most $d_0 - 1$ sign changes (between two sample points of opposite sign $\sin(d_0 \theta)$ has a zero in $(0, \pi)$, and it has $d_0 - 1$ of them). By Laguerre's extension of Descartes' rule (an exponential sum $\sum c_i e^{\beta_i t}$ has at most as many real zeros, with multiplicity, as its coefficient sequence has sign changes; E. Laguerre, *J. Math. Pures Appl.* (3) 9 (1883); G. Pólya, G. Szegő, *Problems and Theorems in Analysis II*, Part V, Ch. 1; proved in [1D, Lemma 3.1]) g has at most $d_0 - 1$ real zeros; it has a zero of multiplicity exactly $d_0 - 1$ at 0 (a single path of length d_0), so none in $(0, \infty)$. Likewise g' has a zero of multiplicity $d_0 - 2$ at 0 and at most one more; g attains its maximum at some $t_c > 0$, so this zero exists, is simple, and $g''(t_c) < 0$. For $d_0 = 1$ all $c_m = \sin^2 \theta_m > 0$. The general statement (any target) is [UN, Thm 4.3(ii)] (variation diminishing for the heat flow on a path). $\square$

Theorem 5.3 (discrete time, nonnegative spectrum; [1D, Thm 3.5], [UN, Thm 4.3(i)]). *Let $q \leq q_N^\perp$ (in particular $q \leq \frac{1}{2}$) and $d_0 \geq 1$. There is $t^* \geq d_0$ with*

$$0 = f(1) = \dots = f(d_0 - 1) < f(d_0) < \dots < f(t^*), \quad f(t^*) \geq f(t^* + 1) > f(t^* + 2) > \dots;$$

so f_{x_0} is unimodal with at most two modes, which are then adjacent. If $q < q_N^\dagger$ and $d_0 \geq 2$, the real-variable increment $\Psi(t) = \Phi(t+1) - \Phi(t)$, $\Phi(t) = \sum_m c_m \lambda_m^{t-1}$, has exactly one zero t_0 in $(d_0 - 1, \infty)$, and every mode lies in $[t_0, t_0 + 1]$. The case $q = q_N^\dagger$ of the strict statement is marked † . For every start and every target on a segment the PMF is unimodal for $q \leq \frac{1}{2}$ [UN, Thm 4.3(i)].

Proof. $q < q_N^\dagger$. All $\lambda_m > 0$; Φ is an exponential sum with strictly decreasing exponents $\ln \lambda_m$ and coefficients c_m/λ_m with at most $d_0 - 1$ sign changes, and so is Ψ (coefficients $c_m(\lambda_m - 1)/\lambda_m$). If $d_0 = 1$, $\Psi < 0$. If $d_0 \geq 2$, Ψ vanishes at $t = 1, \dots, d_0 - 2$, so by Laguerre's rule it has at most one further real zero; $\Psi(d_0 - 1) = \Phi(d_0) > 0$ and Ψ is negative at large integers. Hence exactly one zero $t_0 > d_0 - 1$, simple, with $\Psi > 0$ on $[d_0 - 1, t_0)$, $\Psi < 0$ on (t_0, ∞) ; take $t^* = \lceil t_0 \rceil$.

$q = q_N^\dagger$. Then $\lambda_{N-1} = 0$ and $\lambda_1 > \dots > \lambda_{N-2} > 0$, so for integers $t \geq 2$, $f(t) = \frac{2q}{2N-1} \Phi'(t)$ with the $(N-2)$ -term sum $\Phi'(t) = \sum_{m < N-1} c_m \lambda_m^{t-1}$; put $\Psi'(t) = \Phi'(t+1) - \Phi'(t)$ (real t). Its coefficients are, up to the negative factors $(\lambda_m - 1)/\lambda_m$, a subsequence of (c_m) , so Ψ' has at most $d_0 - 1$ real zeros with multiplicity. $c_1 = \sin \theta_1 \sin(d_0 \theta_1) > 0$ because $0 < d_0 \theta_1 < \frac{\pi}{2}$, so $\Psi' < 0$ for large t , and $f(t+1) - f(t) = \frac{2q}{2N-1} \Psi'(t)$ for integers $t \geq 2$. If $d_0 = 1$, all $c_m > 0$, $\Psi' < 0$, and $f(2) = \frac{q}{2}(1-q) < f(1) = \frac{q}{2}$ ($N \geq 3$): $t^* = 1$. If $d_0 = 2$, Ψ' has at most one real zero and is negative for large t ; with $f(2) = (\frac{q}{2})^2 > 0 = f(1)$ this gives the pattern. If $d_0 \geq 3$, Ψ' vanishes at the $d_0 - 3$ integers $2, \dots, d_0 - 2$, and $\Psi'(d_0 - 1) = \frac{2N-1}{2q} f(d_0) > 0$; since $\Psi' < 0$ for large t , the total multiplicity of its zeros in $(d_0 - 1, \infty)$ is odd and at most $(d_0 - 1) - (d_0 - 3) = 2$, hence 1: one simple zero t_0 , and $t^* = \lceil t_0 \rceil$ as before. (A 60-digit check of all strict patterns at $q = q_N^\dagger$, $3 \leq N \leq 25$, every start, $t \leq 8N^2$, is `code/msc_rigorous_1d/38_qN_strict.py`.) $\square$

Theorem 5.4 (log-concavity; [1D, Thm 3.6]). *From the reflecting end ($x_0 = 1$): if $q \leq \frac{2}{3}$, the PMF is log-concave with support $\{t \geq N - 1\}$; in continuous time the density is log-concave for every q .*

Proof (outline). With $K_t(x) = (Q^t)_{x, N-1}$ and $M_t(x, y) = K_t(x)K_{t-1}(y) - K_{t-1}(x)K_t(y)$, one has $f(t+1)^2 - f(t)f(t+2) = (\frac{q}{2})^3 M_t(1, 2)$, and $M_{t+1}(x, y) = \sum_{z < w} Q[x, y|z, w] M_t(z, w)$ (Cauchy–Binet for 2×2 minors). Q is totally positive of order two iff its only possibly negative contiguous minor $(1 - q)^2 - \frac{q^2}{4}$ is ≥ 0 , i.e. iff $q \leq \frac{2}{3}$; induction gives $M_t \geq 0$. The continuous case follows by letting the move probability tend to 0. Complete proof: [1D, Section 3.4]. $\square$

For $q \leq \frac{1}{2}$ a second proof, from the product formula (convolution of geometric laws preserves log-concavity), is [CE, Thm 28]; Theorem 5.4 supersedes it.

Proposition 5.5 ($N = 3$: exact criterion; [CE, Rem 29(a)]). † *For $d = 1$, $N = 3$, $x_0 = 1$ and every $q \in (0, 1]$, $t \geq 2$:*

$$f(t)^2 - f(t-1)f(t+1) = \frac{q^4}{16} (\det Q)^{t-2}, \quad \det Q = 1 - \frac{3q}{2} + \frac{q^2}{4}.$$

Hence the PMF is log-concave iff $q \leq 3 - \sqrt{5} = 0.7639\dots$, and for $3 - \sqrt{5} < q \leq 1$ log-concavity fails exactly at the odd $t \geq 3$.

Proof. Q is the symmetric 2×2 matrix with rows $(1 - \frac{q}{2}, \frac{q}{2})$, $(\frac{q}{2}, 1 - q)$; its off-diagonal entry is non-zero, so its eigenvalues $x \neq y$ are distinct and, by Cayley–Hamilton, $Q^k = \alpha_k Q + \beta_k I$ with $\alpha_k = (x^k - y^k)/(x - y)$. Hence $f(t) = \frac{q}{2} (Q^{t-1})_{1,2} = \frac{q^2}{4} \alpha_{t-1}$, and the identity $(x^k - y^k)^2 - (x^{k-1} - y^{k-1})(x^{k+1} - y^{k+1}) = (xy)^{k-1}(x - y)^2$ with $k = t - 1$ gives the formula, since $xy = \det Q$. $\det Q \geq 0$ iff $q^2 - 6q + 4 \geq 0$ iff $q \leq 3 - \sqrt{5}$ (for $q \leq 1$); if $\det Q < 0$ the right side is negative exactly when $t - 2$ is odd. $\square$

This agrees with the value $q_{lc}(3) = 0.7639$ found numerically in [1D, Rem 3.7]; for $N = 4$ and $q = \frac{4}{5}$ log-concavity fails at $t = 4$ and at no other $t \leq 400$ (exact; [CE, Rem 29(b)]).

Proposition 5.6 (failure near $q = 1$ from the reflecting end; [1D, Prop 3.9], [UN, Prop 4.4], [CE, Prop 32]). *Let $x_0 = 1$, $N \geq 4$. Then $f(N+1) > f(N)$ for every $q \in (0, 1]$, and $f(N) < f(N-1)$ iff $q > \frac{2N-4}{2N-3}$. Hence for $q > \frac{2N-4}{2N-3}$ the PMF is not unimodal ($f(N-1) > f(N) < f(N+1)$); in particular it is not unimodal for $q = 1$ and any $N \geq 4$. For $N \in \{2, 3\}$ it is unimodal for every q , and for $N = 4$ it is unimodal iff $q \leq \frac{4}{5}$.*

Proof (outline). Path counting: $f(N-1) = (\frac{q}{2})^{N-1}$, $f(N) = (\frac{q}{2})^{N-1}[\eta_1 + (N-2)\eta]$ with $\eta = 1 - q$, $\eta_1 = 1 - \frac{q}{2}$, and $f(N+1) = (\frac{q}{2})^{N-1}[\eta_1^2 + (N-2)\eta_1\eta + \binom{N-1}{2}\eta^2 + (N-2)\frac{q^2}{4}]$; the difference $f(N+1) - f(N)$ is a quadratic in η that is positive on the relevant range [CE, Prop 32], [1D, Prop 3.9] (two separately written hand proofs). For $N = 4$, $q = \frac{4}{5}$, the integer vectors $R_t = (5Q)^t e_1$ are $R_4 = (249, 192, 104)$ and $R_5 = (1131, 898, 488) \leq 5R_4$, so by the cone version of Lemma 4.12 f is nonincreasing from $t = 5$ on, while $f(3) = f(4) = \frac{8}{125} < f(5) = \frac{208}{3125}$: unimodal at $\frac{4}{5}$, hence for $q \leq \frac{4}{5}$ by Theorem 4.10 [UN, Prop 4.4]. $\square$

Theorem 5.7 ($q = 1$: each parity class is unimodal; [1D, Thm 3.10]). *Let $q = 1$ and $d'_0 = 2N - 1 - d_0$. The two sequences $i \mapsto f(d_0 + 2i)$ and $i \mapsto f(d'_0 + 2i)$ are positive, strictly increasing up to some index, and strictly decreasing after it (with at most one equality at the top); $f(t) = 0$ for $t < d_0$, and for $t < d'_0$ with $t \equiv d'_0 \pmod{2}$.*

Computer-assisted theorem 5.8 (eventual total positivity beyond $q = \frac{2}{3}$; [1D, Lemmas 3.12–3.14, Thms 3.15, 3.16]). *From the reflecting end:*

- (a) *for $q = \frac{4}{5}$ and every $N \geq 4$, Q^k is TP_2 for all $k \geq 10$; the PMF is log-concave iff $N = 2$ or $N \geq 5$, and unimodal for every N ;*
- (b) *for $q = \frac{17}{20}$ (every $N \geq 5$, $k \geq 18$) and $q = \frac{9}{10}$ (every $N \geq 7$, $k \geq 40$) Q^k is TP_2 ; the PMF is log-concave iff $N = 2$ or $N \geq 12$ (resp. $N \geq 33$) and unimodal iff $N \notin \{4, 5\}$ (resp. $N \notin \{4, \dots, 8\}$);*
- (c) *consequently, with $q^*(N, 1)$ the unimodality threshold of Theorem 4.10: $q^*(N, 1) \geq \frac{4}{5}$ for all N , $\geq \frac{17}{20}$ for $N \geq 6$, $\geq \frac{9}{10}$ for $N \geq 9$; the bound $\frac{4}{5}$ is optimal for a statement valid for all N (Proposition 5.6, $N = 4$).*

Proof (outline). Products of TP_2 matrices are TP_2 (Cauchy–Binet); for a banded matrix with positive band it suffices to check the contiguous minors; and the minors of Q^k on N sites coincide with minors on a fixed finite chain once $N - 1 \geq 4k$ (locality). So finitely many integer minors of $(10Q)^k$, $10 \leq k \leq 19$, $N \leq 80$ (712 830 contiguous minors; every $k \geq 20$ is a sum of integers in $[10, 19]$) decide (a); similarly (b). Exact integer arithmetic, `code/msc_rigorous_1d/31_tp2_powers.py`; a second, separately written implementation (all 2×2 minors, no band shortcut) is `34_tp2_second_impl.py`. Complete proof: [1D, Section 3.7]. $\square$

The statement “unimodal for every N and every $q \leq \frac{4}{5}$ ” has a second computer-assisted proof, by a different method, through the free propagator (Computer-assisted theorem 6.4 with $d = 1$), and on the finite ranges of Computer-assisted theorem 5.24 a third, by exact certificates.

Conjecture 5.9 ([1D, Rem 3.17(b)]). *For every $q < 1$ the powers Q^k are TP_2 for all $k \geq k_*(q)$, uniformly in N (observed $k_* = 2, 4, 10, 18, 40$ for $q = \frac{7}{10}, \frac{3}{4}, \frac{4}{5}, \frac{17}{20}, \frac{9}{10}$; heuristically $k_* \approx q^2/(2(1-q)^2)$). This would give log-concavity for all $N \geq N_1(q)$.*

Computer-assisted theorem 5.10 (exact one-dimensional thresholds; [UN, Prop 7.3, Thm 7.4]). † *Let $q^*(N)$ be the unimodality threshold (Theorem 4.10).*

- (a) *End to end ($x_0 = 1$): $q^*(2) = q^*(3) = 1$; $q^*(N) = \frac{2N-4}{2N-3}$ for $N = 4$, $N = 20$ and every $22 \leq N \leq 300$; for $5 \leq N \leq 19$ and $N = 21$, $q^*(N) < \frac{2N-4}{2N-3}$, with six-digit brackets $k_N 10^{-6} \leq q^*(N) < (k_N + 1) 10^{-6}$ (e.g. $q^*(21) \in [0.974226, 0.974227]$, whereas $\frac{38}{39} = 0.974358 \dots$).*
- (b) *Centre to corner (N odd, $x_0 = \frac{N+1}{2}$, $M = \frac{N-1}{2}$): $f(M) = (\frac{q}{2})^M$, $f(M+1) = (\frac{q}{2})^M M(1-q)$, $f(M+2) = (\frac{q}{2})^M [(\frac{M+1}{2})(1-q)^2 + M\frac{q}{4}]$ (hand proof). Hence (with a two-step cone certificate at $q = \frac{4}{5}$ for $N = 3$, also by hand) $q^*(3) = \frac{4}{5}$ and, for $N \geq 7$, $q^*(N) \leq \frac{N-3}{N-1}$, for $N = 5$, $q^*(5) \leq \frac{4+\sqrt{2}}{7}$. Exactly: $q^*(5) = \frac{4+\sqrt{2}}{7}$; $q^*(N) = \frac{N-3}{N-1}$ for $N \in \{7, 9, 17, 21\}$ and every odd $23 \leq N \leq 301$; brackets for $N \in \{11, 13, 15, 19\}$.*
- (c) *At $q = \frac{4}{5}$ the centre-to-corner PMF in $d = 1$ is unimodal iff $N \notin \{5, 7, 9, 11\}$ (witness for $N = 5$: $f(2) = \frac{4}{25} > f(3) = \frac{8}{125} < f(4) = \frac{44}{625}$).*

Proof (outline). Upper bounds: Proposition 5.6 and the path counting of (b). Lower bounds and brackets: exact cone certificates (Lemma 4.12) at rational q in integer arithmetic (two separately written engines, Python integers and python-flint integer polynomials, which agree on $4 \leq N \leq 200$ in (a) and odd $7 \leq N \leq 81$ in (b)); $q^*(5)$ is decided in the ring $\mathbb{Z}[\sqrt{2}]$. (c): $N \in \{5, 7, 9\}$ by the path counting of (b), $N = 11$ by the bracket of (b), $N = 3$ and odd $13 \leq N \leq 23$ by the lower bounds of (b), and odd $N \geq 25$ by the log-concavity of the folded free increment, Computer-assisted theorem 6.3(a) of Section 6, with Theorem 4.14(b). Run time: 1.84 h (flint, $4 \leq N \leq 300$), 0.44 h (Python, $N \leq 200$), 0.57 h (centre to corner), wall-clock on a loaded machine. Complete proof: [UN, Section 7.1]. $\square$

Floating-point bisections [1D, Obs 3.20], [UN, Obs 4.5] agree with (a) on their ranges. The heuristic explanation of the exceptional set $\{5, \dots, 19, 21\}$ (a parity zig-zag at the top of the PMF) is [UN, Rem 7.5]; it is not a proof.

Refuted statement 5.11 ([UN, Remark after Conj 10.2]). “In $d = 1$, end to end, $q^*(N) = \frac{2N-4}{2N-3}$ for all $N \geq 20$.” False at $N = 21$: at $q = \frac{38}{39}$, $f(142) > f(143) < f(144)$ (exact integers). The threshold for $N \geq 22$ is the subject of Conjecture 12.1(iii).

5.3 Starts away from the reflecting wall

Proposition 5.12 (failure of discrete unimodality for $x_0 \geq 2$; [1D, Prop 8.8], [UN, Prop 7.3 and the remark after it]). [†] Let $N \geq 3$, $2 \leq x_0 \leq N-1$, $d_0 = N - x_0$, $q \in (0, 1]$. Then

$$f(d_0) = \left(\frac{q}{2}\right)^{d_0}, \quad f(d_0 + 1) = \left(\frac{q}{2}\right)^{d_0} d_0 (1 - q), \quad f(d_0 + 2) = \left(\frac{q}{2}\right)^{d_0} \left[\frac{d_0(d_0+1)}{2} (1 - q)^2 + \frac{d_0}{4} q^2 \right].$$

Consequently $f(d_0) > f(d_0 + 1) < f(d_0 + 2)$, so f is not unimodal, whenever $q > q_{\text{fail}}(d_0)$, where $q_{\text{fail}}(1) = \frac{4}{5}$, $q_{\text{fail}}(2) = \frac{4+\sqrt{2}}{7} = 0.77345\dots$, $q_{\text{fail}}(d_0) = 1 - \frac{1}{d_0}$ ($d_0 \geq 3$). Hence $q_N^{\downarrow} \leq q^*(N, x_0) \leq q_{\text{fail}}(N - x_0)$.

Proof. A path of length t from x_0 to its first visit of N has r right moves, l left moves and s rests with $r - l = d_0$, $r + l + s = t$. For $t = d_0$ there is one path. For $t = d_0 + 1$, one rest at one of the d_0 sites $x_0, \dots, N-1$, all ≥ 2 , so the rest has probability $1 - q$. For $t = d_0 + 2$: either two rests (a multiset of two of these d_0 sites, $\frac{d_0(d_0+1)}{2}$ choices, weight $(1 - q)^2$), or one left move from one of these sites (a genuine move, since all are ≥ 2) followed at once by the right move back, d_0 paths of extra weight $(\frac{q}{2})^2$. This gives the three values. $f(d_0 + 1) < f(d_0)$ iff $d_0(1 - q) < 1$, and with $r := 1 - q$, $(\frac{q}{2})^{-d_0} (f(d_0 + 2) - f(d_0 + 1)) = \frac{d_0}{4} [(2d_0 + 3)r^2 - 6r + 1]$. For $d_0 = 1$ the bracket is $(5r - 1)(r - 1)$, positive iff $r < \frac{1}{5}$, and $f(2) < f(1)$ always. For $d_0 = 2$, $7r^2 - 6r + 1 > 0$ iff $r < \frac{3-\sqrt{2}}{7}$ or $r > \frac{3+\sqrt{2}}{7}$; with $r < \frac{1}{2}$ this is $q > \frac{4+\sqrt{2}}{7}$. For $d_0 = 3$ the bracket is $(3r - 1)^2 > 0$ for $q \neq \frac{2}{3}$. For $d_0 \geq 4$ its discriminant $8(3 - d_0)$ is negative. A unimodal sequence cannot have $f(d_0) > f(d_0 + 1) < f(d_0 + 2)$. Finally $q_N^{\downarrow} \leq q^*$ by Theorem 5.3 and $q^* \leq q_{\text{fail}}$ because the unimodal set is $(0, q^*]$ (Theorem 4.10). $\square$

The two notes contain this argument in two separately written forms (general $x_0 \geq 2$ in 1D; the centre start and general pairs in UN); for $N = 3$, $x_0 = 2$ unimodality holds iff $q \leq \frac{4}{5}$ [UN, Prop 7.3(a)] (exact cone certificate at $q = \frac{4}{5}$).

Computer-assisted proposition 5.13 (q_{fail} is not the threshold; [1D, Obs 8.9], [UN, remark after Prop 7.3]). [†] For $d_0 = 5$, $q = \frac{4}{5} = q_{\text{fail}}(5)$ and $8 \leq N \leq 13$ (so $x_0 \geq 3$) the PMF is not unimodal (a dip at $t = 10$, $f(9) > f(10) < f(11)$, exact rational arithmetic). At $q = \frac{4}{5}$, among all 66 starts with $N \leq 12$ exactly 29 are not unimodal, all with $x_0 \geq 2$ and $d_0 \in \{2, 3, 4, 5\}$; the other 37 are unimodal (exact cone certificates). In particular $N = 7$, $x_0 = 2$ is unimodal.

Numerical observation 5.14 ([1D, Obs 8.9]). An exact scan over $3 \leq N \leq 13$, $2 \leq x_0 \leq N - 1$, $q \in \{\frac{3}{5}, \frac{2}{3}, \frac{3}{4}, \frac{4}{5}, \frac{9}{10}, 1\}$, $t \leq 6N^2$ finds a certified failure of unimodality in 171 of the 396 cases: the 165 cases with $q > q_{\text{fail}}(d_0)$ and the six cases of Computer-assisted proposition 5.13. In the other 225 cases no failure was found up to $t = 6N^2$ (a finite search, which proves nothing).

5.4 The continuum limit and the constant c

Let $y_k = k\pi/4$, $\sigma_k = (-1)^{(k-1)/2}$ for odd k , and

$$G(u) = \frac{\pi}{2} \sum_{k \text{ odd}} \sigma_k k e^{-k^2 \pi^2 u/8} \quad (u > 0),$$

the density of the first time a standard Brownian motion started at 0 leaves $(-1, 1)$, i.e. of the passage time of reflected Brownian motion from 0 to 1 (Laplace transform $1/\cosh \sqrt{2}s$). The scaled lattice density is $G_N(u) = \frac{L^2}{q} g(\frac{L^2 u}{q})$ (independent of q).

Theorem 5.15 (the limit density; [1D, Thms 4.2, 4.3, 8.3, Rem 8.4]). $G_N^{(j)} \rightarrow G^{(j)}$ uniformly on $[u_0, \infty)$ for every $u_0 > 0$ and $j \geq 0$. $G > 0$, $\log G$ is concave, G' has exactly one zero c in $(0, \infty)$, and $G''(c) < 0$. (Uniqueness and non-degeneracy also follow by hand, without numerical evaluation, from Descartes' rule on the lattice, [1D, Thm 8.3 with $\xi = 0$].)

Computer-assisted theorem 5.16 (the constant; [1D, Thm 4.6], [CE, Thm 23]). With the 80-digit decimal c_0 of `data/msc_rigorous_1d/03_constants.json`, $|c - c_0| < 10^{-70}$, and

$$c = 0.3332842654949478987485269165434424210970365590711318852610059370862\dots$$

Moreover $G(c) = 0.92506494218414348715254834\dots$, $G''(c) = -12.514127592957827401576602\dots$, and

$$\kappa := \frac{3}{4} + \frac{c}{6} \frac{G'''(c)}{G''(c)} = 0.086273779378239017181791\dots, \quad \rho := -\frac{c}{2} \frac{G'''(c)}{G''(c)} = 1.9911786618652829484546\dots$$

A separately written proof of uniqueness (via the dual theta series, Jacobi's identity and the modular transformation of η) and a 29-digit enclosure of $c = \kappa_1 = 2\tau^*$ are [CE, Thm 23]; the two enclosures agree.

Proof (outline). `code/msc_rigorous_1d/03_constant_enclosure.py`: $G'(c_0 \mp 10^{-70}) \geq 0$ in 400-bit Arb ball arithmetic, the series tails bounded by an explicit geometric bound [1D, Lemma 4.1]; Theorem 5.15 turns the sign change into the enclosure. $\square$

By the method of images the two leading images give the Lévy density with maximiser $\frac{1}{3}$; the next images are of relative size $3e^{-12}$ at $u = \frac{1}{3}$, which is why $c = \frac{1}{3} - 4.9068 \dots \times 10^{-5}$ [1D, Rem 4.5].

5.5 The mode

Computer-assisted theorem 5.17 (continuous time; [1D, Thm 5.7, Cor 5.8]). *Let t_c be the continuous-time mode from the reflecting end and $u_N = qt_c/L^2$. For every $N \geq 3$,*

$$\left| u_N - c + \frac{\kappa}{L^2} \right| < \frac{1.78}{L^4} \quad (1.54 \text{ for } N \geq 201), \quad c - \frac{0.3614}{L^2} < u_N < c - \frac{\kappa}{L^2},$$

and $t_c/\mathbb{E}T = c + (\frac{c}{4} - \kappa)L^{-2} + \vartheta L^{-4}$ with $\frac{c}{4} - \kappa = -0.0029527130 \dots$ and $|\vartheta| \leq 1.86$. $\dagger$ The limit $\kappa_2 = \lim_N L^4(u_N - c + \kappa L^{-2})$ exists and equals $-\Gamma_0/G''(c)$ with

$$\Gamma_0 = \frac{1}{2}\kappa^2 G''' - \kappa(\frac{11}{12}G''' + \frac{c}{6}G''''') + \frac{113}{480}G''' + \frac{49c}{360}G'''' + \frac{c^2}{72}G^{(5)} \quad (\text{at } c), \quad \kappa_2 = -1.5308900198375030446 \dots$$

(the existence proof and the closed form were written after the internal checks; the bounds were checked in them).

Proof (outline). A lattice expansion with certified remainder, $G_N^{(j)}(v) = G^{(j)}(v) + L^{-2}H_j(v) + L^{-4}R$ with $H_1 = \frac{3}{4}G''' + \frac{v}{6}G''''$ and R enclosed in ball arithmetic uniformly in $L \geq L_0$ [1D, Prop 5.4] (by hand, with explicit constants), is inserted at $u = c - \kappa L^{-2} \pm \kappa L^{-4}$; Taylor's formula at c and the unimodality of Theorem 5.2 give the window for $N \geq N_0$ (script 12), and $3 \leq N \leq 300$ are checked one by one in 160-bit ball arithmetic (script 13). The limit κ_2 : the remainder terms converge by dominated convergence (Tannery's theorem) uniformly in $|s| \leq 1.78$, and the limit is evaluated in closed form; its value is enclosed in 256-bit ball arithmetic (script 37) and lies inside the certified window $[-1.5309330, -1.5308464]$ of the row $N_0 = 2001$. Complete proof: [1D, Section 5]. $\square$

Computer-assisted theorem 5.18 (discrete time; [1D, Thms 6.1, 6.3, Prop 6.2, Cor 6.4]). *Let $x_0 = 1$ and (N, q) satisfy (a) $N \geq 3$, $q \leq \frac{4}{5}$; or (b) $N \geq 7$, $q \leq \frac{17}{20}$; or (c) $N \geq 10$, $q \leq \frac{9}{10}$. Then f is unimodal and every mode satisfies*

$$\tau - E < t^* < \tau + 1 + E, \quad \tau = \frac{cL^2 - \kappa}{q} + 1 - \rho, \quad E = \frac{1.78}{qL^2}$$

(1.68 in place of 1.78 for $N \geq 41$, $q \leq \frac{4}{5}$; 1.61 for $N \geq 61$). Consequently $|qt^*/L^2 - c| < (0.0863 + 0.9912q)L^{-2} + 1.78L^{-4}$; $c - qt^*/L^2 \geq 0.0774L^{-2} - 1.78L^{-4} > 0$ for $N \geq 6$ (the exponent 2 is optimal); $|t^*/\mathbb{E}T - c| \leq (0.003 + 0.992q)L^{-2} + 2.2L^{-4}$; and if $\text{dist}(\tau, \mathbb{Z}) \geq E$, the mode is unique and $t^* = \lceil \tau \rceil$. The restrictions $N \geq 7$, $N \geq 10$ in (b), (c) cannot be dropped: for $N = 6$, $q = 0.828$ the mode is $11 < \tau - E = 11.0097 \dots$, and for $N = 9$, $q = 0.887$ it is $26 < \tau - E = 26.031 \dots$ (exact rational computation).

Proof (outline). Unimodality: Computer-assisted theorem 5.8. Window: the expansion of the increment $\frac{L^4}{q^2}(f(t+1) - f(t)) = G'(v) + L^{-2}H(v) + L^{-4}R$, $v = q(t-1)/L^2$, $H = \frac{3}{4}G''' + \frac{v}{6}(1-3q)G''''$, certified on 2400 (resp. 2700) q -intervals uniformly in $L \geq L_0$, first with per-interval constants (script 14) and then with the single constant of the statement on every interval (script 35, 0 failing intervals; a re-run with a uniform constant by separately written code within the same project gave the same result); $3 \leq N \leq 60$ by bisection in q (scripts 15, 32). Complete proof: [1D, Section 6]. $\square$

For $q = \frac{4}{5}$ the exact formula reads $t^* = \lceil \frac{5}{4}cL^2 - 1.09902 \dots \rceil$, and for $q = \frac{1}{2}$, $t^* = \lceil 2cL^2 - 1.16372 \dots \rceil$, valid for every $N \geq 3$ with $\text{dist}(\tau, \mathbb{Z}) \geq E$ [1D, Remark after Cor 6.4]. On the finite ranges of Computer-assisted theorem 5.24 the modes are certified case by case (rules $t^* = \lfloor cL^2/q - 0.0997 \rfloor$ and $\lfloor cL^2/q - 0.164 \rfloor$ for $N \geq 10$). On these ranges ($3 \leq N \leq 1500$ and $N = 2000$ for $q = \frac{4}{5}$; $3 \leq N \leq 1000$ for $q = \frac{1}{2}$; $E = 1.78/(qL^2)$, resp. $1.68/(qL^2)$ for $N \geq 41$) the hypothesis $\text{dist}(\tau, \mathbb{Z}) \geq E$ holds in 1495 of 1499 cases for $q = \frac{4}{5}$ (all except $N \in \{4, 9, 12, 13\}$) and in 989 of 998 cases for $q = \frac{1}{2}$ (all except $N \in \{3, 4, 6, 7, 9, 10, 12, 13, 59\}$), and in each of them the formula gives the certified mode (on the sub-ranges $3 \leq N \leq 600$, $N \in \{800, 1000\}$ and $3 \leq N \leq 300$ used in [1D, Remark after Cor 6.4]: 596 of 600 and 289 of 298 cases). For the remaining N (τ within E of an integer) the window of Computer-assisted theorem 5.18 contains more than one integer and does not by itself single out the mode.

Computer-assisted theorem 5.19 (the simple random walk $q = 1$; [1D, Thm 7.4]). *Let $q = 1$, $x_0 = 1$, $N \geq 3$. Every global maximiser satisfies $t^* \equiv N-1 \pmod{2}$ and $\tau_1 - \frac{K}{L} < t^* < \tau_1 + 2 + \frac{K}{L}$ with $K = 10.3$ ($K = 0.031$ for $N \leq 600$), where $\tau_1 = cL^2 - bL + 1 + s_2$, $b = 0.33250858299642786\dots$, $s_2 = -2.2503062886311552\dots$. There are at most two maximisers (t^* and $t^* + 2$), which happens for $N \in \{5, 6, 8, 9\}$ and for no other $N \leq 600$. Hence $L(c - t^*/L^2) \rightarrow b \neq 0$: for $q = 1$ the scaled mode converges to c at the exact rate N^{-1} .*

Computer-assisted theorem 5.20 (global maximisers for $\frac{2}{3} \leq q < 1$ without unimodality; [1D, Thm 7.7, Cor 7.8]). *Let $N \geq 601$, $q \in (0, 1)$, $\chi(q) = \frac{1}{2}$ for $q \leq \frac{4}{5}$, $\chi(q) = \min\{\frac{1}{2}, -\ln(2q-1)/q\}$ for $q > \frac{4}{5}$, and assume $4(\frac{\pi}{2})^3 L^8 \exp(-0.15\chi(q)L^2) \leq 10^{-12}$ (true for all $N \geq 601$ if $q \leq 0.999$, for $N \geq 2001$ if $q \leq 0.9999$). Then every global maximiser satisfies $\tau - E < t^* < \tau + 1 + E$ with $E = 1.56/(qL^2)$; there are at most two, and they are adjacent. Consequently, for every fixed $q \in (0, 1)$, $qt^*/L^2 \rightarrow c$ and $t^*/\mathbb{E}T \rightarrow c$ as $N \rightarrow \infty$ (error $O(N^{-2})$ for $q < 1$, $-b/L + O(N^{-2})$ for $q = 1$; also in continuous time).*

The hypothesis is essentially $(1-q)L^2 \gtrsim 27 \ln L + 100$ and is not an artefact: for $1-q \lesssim L^{-2}$ the mode follows the $q = 1$ law (e.g. $N = 10$, $q = 0.99$: $t^* = 27$, the $q = 1$ mode, while $\tau = 29.30\dots$) [1D, Rem 7.10]. The crossover regime is open (Section 12).

Refuted statement 5.21 (the closed form in $d = 1$; [1D, Rem 7.9], [CE, Thms 21(c), 43(c)]). *“ $t^* \approx t_{\text{cf}}$ in every dimension.” False for $d = 1$: there $X = (1 - \cos \frac{\pi}{N})N(N-1) \rightarrow \frac{\pi^2}{2}$, so $t_{\text{cf}}/\mathbb{E}T \rightarrow \ln(\pi^2)/(1 + \frac{\pi^2}{2}) = 0.385768\dots$, whereas $t^*/\mathbb{E}T \rightarrow c = 0.333284\dots$ (Computer-assisted theorem 5.20); the closed form overestimates the mode by a factor tending to $1.15747\dots$. On finite ranges (computer-assisted, CE): $0.1577t^* \leq t_{\text{cf}} - t^* \leq 0.2251t^*$ for $q = \frac{4}{5}$, $10 \leq N \leq 1500$ and $N = 2000$; $0.1579t^* \leq t_{\text{cf}} - t^* \leq 0.2282t^*$ for $q = \frac{1}{2}$, $10 \leq N \leq 1000$. In $d = 1$ all poles are of order L^{-2} and all contribute at the mode.*

5.6 General start

For $0 \leq \xi < 1$ let $G_\xi(u) = \frac{\pi}{2} \sum_{k \text{ odd}} \sigma_k k \cos \frac{k\pi\xi}{2} e^{-k^2\pi^2 u/8}$ ($G_0 = G$).

Theorem 5.22 (limit density and limit theorem for a general start; [1D, Lemmas 8.1, 8.2, Thms 8.3, 8.5]). *$G_\xi > 0$ is real-analytic and has exactly one critical point $c(\xi)$, a non-degenerate maximum; G''_ξ has exactly two sign changes; $\xi \mapsto c(\xi)$ is real-analytic on $[0, 1)$, $c(0) = c$. If $(x_0 - \frac{1}{2})/L \rightarrow \xi$, then $qt_c/L^2 \rightarrow c(\xi)$ in continuous time (every q), and $qt^*/L^2 \rightarrow c(\xi)$ in discrete time for $q \leq \frac{1}{2}$; $\text{mode}/\mathbb{E}T \rightarrow c(\xi)/(1 - \xi^2)$.*

The proof transfers the sign-change bounds of Laguerre’s rule from the lattice densities to the limit (Lemma 8.2 of 1D) and uses the image representation of G_ξ (Poisson summation) for the behaviour at 0; no numerical evaluation is used.

Computer-assisted theorem 5.23 (explicit law for starts in the half next to the wall; [1D, Prop 8.6, Thm 8.12, Cor 8.14]). *Let $N \geq 3$, $\xi_N = (x_0 - \frac{1}{2})/L \leq \frac{1}{2}$, $c_N = c(\xi_N)$, $\rho_N = -\frac{c_N}{2} G'''_{\xi_N}/G''_{\xi_N}$ and $\kappa_N = \frac{1}{2} - \frac{\rho_N}{3}$ (at c_N). Then $|qt_c/L^2 - c_N + \kappa_N/L^2| < 3.5/L^4$ in continuous time, and in discrete time with $q \leq \frac{1}{2}$ every mode satisfies $\tau - E < t^* < \tau + 1 + E$ with $\tau = (c_N L^2 - \kappa_N)/q + 1 - \rho_N$, $E = 3.5/(qL^2)$. Uniformly, $1.7206 < \rho_N < 2.5171$, $-0.3391 < \kappa_N < -0.0735$, $|qt^*/L^2 - c(\xi_N)| < 0.44L^{-2} + 3.5L^{-4}$ ($q \leq \frac{1}{2}$). Values: $c(0.5) = 0.08334564332977\dots$, $c(0.7) = 0.030000000000000000000000002859\dots$; for $\xi \geq \frac{1}{2}$, $c(\xi) \approx (1 - \xi)^2/3$ to relative error 1.5×10^{-4} at the tabulated points.*

(The certified constant 3.5 was reached by adaptive bisection; the largest value of the quantity it bounds found in tests is 3.419, at $N = 6$, $x_0 = 3$ [1D, Rem 8.13].)

5.7 Certified modes on finite ranges

Computer-assisted theorem 5.24 (exact modes in $d = 1$; [CE, Thms 12, 13, 30, 39, 40, 45], [1D, Prop 3.18]). *For $q = \frac{4}{5}$ and every $2 \leq N \leq 1500$ and $N = 2000$, and for $q = \frac{1}{2}$ and every $2 \leq N \leq 1000$, the PMF from the reflecting end is strictly unimodal with the certified integer mode, except in three cases with an exact tie: $(q, N) = (\frac{4}{5}, 4)$: $f(3) = f(4) < f(5)$, mode 5; $(\frac{1}{2}, 3)$: maximum at $t \in \{3, 4\}$; $(\frac{1}{2}, 4)$: maximum at $t \in \{7, 8\}$. For $10 \leq N \leq 1500$ and $N = 2000$ ($q = \frac{4}{5}$): $-1.0989 < t^* - cL^2/q < -0.0997$, hence $t^* = \lfloor cL^2/q - 0.0997 \rfloor$; for $10 \leq N \leq 1000$ ($q = \frac{1}{2}$): $-1.160 < t^* - cL^2/q < -0.164$. For $q = 1$ and $4 \leq N \leq 400$ the PMF has at least two strict local maxima (up to 164054 at $N = 400$), and the maximum is attained twice exactly for $N \in \{5, 6, 8, 9\}$.*

The ranges $N \leq 600$ ($q = \frac{4}{5}$), $N \leq 300$ ($q = \frac{1}{2}$), $N \leq 200$ ($q = 1$) rest on Python integers only (trusted base (T1) of Section 9); the extended ranges rest in addition on a C program (trusted base (T5)). For $q = \frac{4}{5}$, $N \leq 600$, a second proof with a different method (IEEE-754 error bars plus Arb at the flat points) is [1D, Prop 3.18]; the two agree on all 598 modes $3 \leq N \leq 600$ (script `code/msc_rigorous_1d/22_crosscheck_t5.py`). All of this is consistent with Computer-assisted theorem 5.18, which covers every N .

6 Unimodality in every dimension

6.1 The three geometries for all N and d

Theorem 6.1 (CC, C2M, M2C: all N , all d ; [UN, Thm 4.12]). *Let $d \geq 1$, $N \geq 2$, and let (x_0, a) be one of CC, C2M, M2C (the last two for odd N), or an image of these under a symmetry of the box. Then (i) for every rate $q > 0$ the continuous-time density g_q is unimodal with a unique mode; (ii) for every $q \in (0, q_N^\square]$, $q_N^\square = 1/(1 + \cos \frac{\pi}{N}) > \frac{1}{2}$, the PMF f_q is unimodal. For $N = 2$, $q_2^\square = 1$: the CC PMF of $\{1, 2\}^d$ is unimodal for every $q \in (0, 1]$.*

Proof. One-dimensional factor: for CC the coordinate walk $K_q = (1 - q)I + qP^{(1)}$ on $\{1, \dots, N\}$ is a birth-death chain between its two ends; for C2M and M2C the folded chain $|X - c|$ on $\{0, \dots, \frac{N-1}{2}\}$ is one, and $K_q^n(c, N) = K_q^n(1, c)$ by symmetry. Its eigenvalues are among $1 - q(1 - \cos \frac{k\pi}{N})$, all ≥ 0 when $q \leq q_N^\square$. For a birth-death chain the end-to-end propagator is π_M times the distribution function of a sum of independent exponential (for $q \leq q_N^\square$ geometric) variables [UN, Lemma 4.9] (Cramer’s rule, as in Theorem 5.1); such distribution functions and their densities (mass functions) are log-concave [UN, Lemma 4.10(a)]. (i) $\mathcal{L} = \frac{1}{d} \sum_i \mathcal{L}_i^{(1)}$ is a Kronecker sum, so $u(t) = h(t)^d$ with h the one-dimensional propagator at rate q/d ; $u' = dh^{d-1}h'$ is positive and log-concave. Apply Theorem 4.14(a). (ii) $P_q = \frac{1}{d} \sum_i K_q^{(i)}$ with commuting summands; the multinomial theorem gives $d^{t-1}\varphi(t) = (\delta h \star h \star \dots \star h)_{t-1}$ with the binomial convolution $(a \star b)_n = \sum_k \binom{n}{k} a_k b_{n-k}$ and $\delta h(n) = h(n+1) - h(n)$. For $q \leq q_N^\square$, h and δh are log-concave (Lemma 4.10(a)(iii) of UN), the binomial convolution preserves log-concavity (Davenport-Pólya, *Canad. J. Math.* 1 (1949); proved, with leading zeros allowed, in [UN, Lemma 4.10(b)]), and so does multiplication by $d^{-(t-1)}$. So φ is log-concave and Theorem 4.14(b) applies. $\square$

In probabilistic terms $N^d u(t) = \mathbb{P}(\max(\tau_1, \dots, \tau_d) \leq t)$ with independent hypoexponential τ_i : the first-passage density is the density of this maximum smoothed by the escape kernel. For CC in continuous time a second proof, from the decomposition $\mu_1 T \stackrel{d}{=} U + \mu_1 T_\pi$ of Proposition 8.1 and the log-concavity of the density of U , is [AS, Thm 1.11] (Theorem 8.2 below); for CC in discrete time with $q \leq \frac{1}{2}$ a second proof, which relies on the published closure of ultra-log-concave sequences under convolution (D. W. Walkup, *J. Appl. Probab.* 13 (1976), Thm 1; T. M. Liggett, *J. Combin. Theory Ser. A* 79 (1997), Thm 2; the theorem numbers are those given by O. Johnson, *Stoch. Proc. Appl.* 117 (2007), and a proof is A. Marsiglietti, J. Melbourne, *Discrete Comput. Geom.* 71 (2024), Cor. 3.4 of arXiv:2004.12005v4), is [AS, Thm 7.4]. Theorem 6.1 is proved by hand without that citation.

Corollary 6.2 (where the continuous-time mode is; [UN, Cor 4.13]). *In the situation of Theorem 4.14(a) with $D \geq 2$, let m_u be the maximiser of u' and, for $a' > 0$, $m(a') := \inf\{t \geq m_u : u'(t) - a' \int_0^t e^{-a'(t-s)} u'(s) ds \leq 0\}$. Then $m_u < t^* \leq m(q\alpha_1) \leq m(q/M_{\max})$, where α_1 is the smallest eigenvalue of $\mathcal{L}_a$ and $M_{\max} = \max_x \mathbb{E}_x T$ at $q = 1$.*

(Numerically the upper bound exceeds the mode by 4.7–24.9% for the CC pair, $d = 2, 3$, $8 \leq N \leq 64$, and is cruder for small N and in $d = 1$; these percentages are floating-point illustrations, not theorems.)

6.2 Discrete time up to the value $q = \frac{4}{5}$

Let $h_q(n) = K_q^n(1, N)$ (CC) and $\tilde{h}_q(n) = K_q^n(1, c)$ (centre geometries), $\delta h_q(n) = h_q(n+1) - h_q(n)$, $q_1(N) := 1/(1 + (2(N-1))^{-1/2})$, and let $\tilde{q}_1(N)$ be the analogous threshold of the folded chain (the value of q at which e_2 of its spectrum vanishes, [UN, Thm 9.7(d)]).

Computer-assisted theorem 6.3 (one-dimensional log-concavity of the free increment; [UN, Prop 9.6, Thms 9.7, 9.11, Cor 9.8, Lemmas 9.9, 9.10]). *(a) For every $q \in (0, \frac{4}{5}]$: δh_q is log-concave for every $N \geq 9$, and $\delta \tilde{h}_q$ for every odd $N \geq 25$; at $q = \frac{4}{5}$ these ranges are exact. (b) For $4 \leq N \leq 501$, δh_q is log-concave iff $q \leq q_1(N)$; for odd $7 \leq N \leq 1001$, $\delta \tilde{h}_q$ is log-concave iff $q \leq \tilde{q}_1(N)$.*

Proof (outline). Up to a geometric factor δh is $W_n = h_{n-N+2}(\lambda_1, \dots, \lambda_{N-1})$, the complete homogeneous symmetric polynomial of $\lambda_k = y + 2 \cos \frac{k\pi}{N}$, $y = 2(1 - q)/q$. Its log-concavity for $n \geq n_{\text{tail}}(N) = 1 + \frac{N^2}{8}(10 \ln N - 8.43)$ follows from the two leading terms (by hand, Prop. 9.6); below n_{tail} it is certified in ball arithmetic and cross-checked in exact integer arithmetic for $N \leq 1001$ (Thm 9.7). For $N \geq 992$ the spectrum splits into residue classes modulo $G = \lfloor N/16 \rfloor$, each a “twisted group” of 15–17 points of a compact two-parameter family; h of the whole spectrum is the ordinary convolution of the h ’s of the groups (Lemma 9.9, by hand), ordinary convolution preserves log-concavity (Lemma 9.2), and every member of the family is log-concave (Computer-assisted Lemma 9.10: Taylor models with Cauchy remainder bounds on

26 parameter boxes, 19 seconds). A second implementation of Lemma 9.10, written separately within the same project (different remainder bound, own tail criterion), certifies the same 26 boxes. “Only if” in (b): $\delta h(N-1)^2 - \delta h(N-2)\delta h(N)$ is a positive multiple of $2(N-1)(1-q)^2 - q^2$. Complete proof: [UN, Section 9]. $\square$

Computer-assisted theorem 6.4 (the value; [UN, Prop 9.1, Prop 9.12, Thm 9.13]). *Let $q \in (0, \frac{4}{5}]$. (i) The CC PMF is unimodal for every $N \geq 2$ and every $d \geq 1$. (ii) The C2M and M2C PMFs are unimodal for every odd $N \geq 3$ and every $d \geq 2$. (iii) $\frac{4}{5}$ is sharp in (i): for $d = 1$, $N = 4$ the PMF is not unimodal for any $q > \frac{4}{5}$ (Proposition 5.6). The hypothesis $d \geq 2$ in (ii) cannot be dropped: in $d = 1$ the M2C PMF at $q = \frac{4}{5}$ is unimodal iff $N \notin \{5, 7, 9, 11\}$ (Computer-assisted theorem 5.10(c)).*

Proof (outline). By the binomial-convolution identity in the proof of Theorem 6.1(ii), log-concavity of the one-dimensional free increment implies that of φ in every dimension (Prop. 9.1); Theorem 6.3(a) and Theorem 4.14(b) give (i) for $N \geq 9$ and (ii) for $N \geq 25$. For the finitely many smaller N blocks of two or three dimensions are log-concave (Computer-assisted Prop. 9.12, 38 cases, ball and exact integer arithmetic) and are combined by the same identity; $N = 2$ is Theorem 6.1(ii) ($q_2^\square = 1$); the remaining cases ($d = 1$, $3 \leq N \leq 8$; centre geometries with $N = 3$, $d \in \{2, 3\}$ and $N \in \{5, 7\}$, $d = 2$) are exact cone certificates (Computer-assisted theorem 6.5(e),(f)). Theorem 4.10 passes from $q = \frac{4}{5}$ to all $q \leq \frac{4}{5}$. Complete proof: [UN, Section 9.5]. $\square$

Computer-assisted theorem 6.4(i) settles, for $q \in \{\frac{4}{5}, \frac{1}{2}\}$ and in the weak sense of unimodality, the conjecture [CE, Conj 46] for every N and d ; strict unimodality for all N (unique maximiser, no plateau except the three exact ties of Computer-assisted theorem 5.24) remains open beyond the certified ranges.

6.3 Beyond $\frac{4}{5}$, and $q = 1$

Computer-assisted theorem 6.5 (finite ranges for every q ; [UN, Thm 7.1]). *(a) $d = 2$, CC: for $2 \leq N \leq 64$ and every $q \in (0, \frac{499}{500}]$ the PMF is unimodal. (b) $d = 3$, CC, $2 \leq N \leq 30$; $d = 4$, $N \leq 8$; $d = 5$, $N \leq 5$: unimodal for every $q \in (0, 1]$. (c) $d = 2$, CC, $q = 1$: for $2 \leq N \leq 64$ the PMF is unimodal except for $N \in \{9, 11, 13, 15, 17, 18, 21, 24, 26, 38, 51, 61\}$, where $f(t^* - 1) > f(t^*) < f(t^* + 1)$ (one dip of one time step next to the maximum). (d) $0.9986 \leq q^*(9) < 0.9987$ for $d = 2$, $N = 9$. (e) C2M and M2C: unimodal for every $q \leq \frac{4}{5}$, $d = 2$, odd $3 \leq N \leq 61$, and $d = 3$, odd $3 \leq N \leq 27$. (f) $d = 1$, CC: unimodal for every $q \leq \frac{4}{5}$, $2 \leq N \leq 60$.*

Proof (outline). Exact integer cone certificates (Lemma 4.12) at one rational q_0 per case, extended to $q \leq q_0$ by Theorem 4.10. Size: (a) 63 runs, up to 7035 steps on 4096 sites, 88 CPU-minutes; (b) $d = 3$: 29 runs, 20 CPU-minutes. Cross-checks with separately written implementations (fraction-based stepper, dictionary-based certificate) in `code/msc_rigorous_unimodality/verify_certificate_code.py`. $\square$

Computer-assisted theorem 6.6 ($q = 1$, extended; [CE, Thms 13, 40]). *For $q = 1$ and the CC pair: in $d = 2$, $2 \leq N \leq 160$ and $N \in \{180, 200\}$, the PMF has two strict local maxima exactly for $N \in \{9, 11, 13, 15, 17, 18, 21, 24, 26, 38, 51, 61, 86, 92, 136, 149\}$ and is strictly unimodal otherwise, except $N = 2$ ($f(2) = f(3)$); in $d = 3$, $2 \leq N \leq 60$, it is strictly unimodal. The four cases $N = 86, 92, 136, 149$ were re-derived with Python integers only.*

The same phenomenon appears in the asymptotics note: $d = 2$, $N = 9$, $q = 1$: $\mathbb{P}(T = 108) = 0.0020122644\dots > \mathbb{P}(T = 109) = 0.0020121967\dots < \mathbb{P}(T = 110) = 0.0020122182\dots$ (exact integers, [AS, Rem 7.19(ii)]).

Refuted statement 6.7 ([UN, Section 8], [CE, Section 9]). “The point-target first-passage law is always unimodal” and “the PMF is unimodal for every $q \in (0, 1]$ ”. False: in discrete time near $q = 1$ (Proposition 5.6, Computer-assisted theorems 6.5(c), 6.6), for starts away from the wall in $d = 1$ (Proposition 5.12), and for general pairs in $d \geq 2$ even in continuous time (Computer-assisted theorem 6.9).

6.4 Explicit windows for the CC pair, every q

Theorem 6.8 (tail and rise; [UN, Thms 5.6, 5.10]). *Let $d \geq 2$, CC, α_1 the smallest eigenvalue of $\mathcal{L}_a$, $\beta_1 = \Lambda_1$. For every $q \in (0, 1]$, $f_q(t+2) < f_q(t+1)$ for every integer $t > T_N(q) := \ln(4\sqrt{n-1}/\alpha_1^2)/\ln \frac{1-q\alpha_1}{1-q\beta_1}$, and $T_N(q) \leq \bar{T}_N(q) = O(N^2 \log N/q)$ explicitly; in continuous time $g'_q < 0$ beyond the analogous time, and g_q is nondecreasing on $[0, t_N]$ with $t_N \geq d(N-2)/(2q)$ ($d \geq 1$, $D \geq 2$).*

Thus for every $q \leq 1$ a possible failure of unimodality of the CC law is confined to an explicit bounded window.

6.5 General pairs

Computer-assisted theorem 6.9 (explicit counter-examples; [UN, Thm 8.1]). (a) $d = 3$, $N = 3$, $x_0 = (3, 2, 2)$, $a = (1, 2, 2)$ (opposite face centres): $g_1(\frac{21}{10}) > g_1(3) < g_1(13)$, and at $q = \frac{1}{2}$, $f(2) = \frac{1}{144} > f(6) = \frac{6577}{995328} < f(25)$. (b) $d = 2$, $N = 8$, $x_0 = (6, 6)$, $a = (3, 4)$: $g_1(\frac{21}{2}) > g_1(\frac{49}{2}) < g_1(\frac{59}{2})$, and at $q = \frac{1}{2}$, $f(19) > f(46) < f(63)$. In both cases neither the density (any rate) nor the PMF (any q) is unimodal.

Proof (outline). $e^t g_1(t) = \sum_n \frac{t^n}{n!} f_1(n+1)$ (identity (1)) with exact rational $f_1(n+1)$; truncation with explicit tail bounds turns each comparison into a comparison of exact rationals (relative width below 10^{-66}). Theorem 4.10 gives all q . $\square$

Computer-assisted theorem 6.10 (exhaustive classification of small boxes; [UN, Thm 8.2]). For $d = 2$, $2 \leq N \leq 12$, and $d = 3$, $2 \leq N \leq 6$, every ordered pair (target up to symmetry) is classified exactly as: unimodal at $q = \frac{1}{2}$; not unimodal at $q = \frac{1}{2}$ but unimodal at some $q = 2^{-k}$, $k \leq 6$; or not unimodal in continuous time. In $d = 2$ every pair is unimodal in continuous time for $N \leq 7$; the smallest bimodal pair is at $N = 8$; there are 54 bimodal pairs in the table, each with a coordinate in which start and target lie strictly on opposite sides of the mid-plane.

Proposition 6.11 (unimodal but not log-concave; [UN, Prop 8.4]). For $d = 2$, $N = 3$, $M2C$ ($x_0 = (2, 2)$, $a = (3, 3)$), $q = \frac{1}{2}$: $f(4) = \frac{67}{2048}$, $f(5) = \frac{481}{16384}$, $f(6) = \frac{3457}{131072}$, so $f(5)^2 < f(4)f(6)$: the PMF is unimodal but not log-concave.

Conjecture 6.12 ([UN, Conjs 10.1, 10.3], [CE, Conj 47]). (a) For the CC pair, $d \geq 2$, in continuous time and in discrete time for $q \leq \frac{1}{2}$, the occupation ratios $\rho_{t+1}(x)/\rho_t(x)$ are coordinatewise nondecreasing in x ; consequently the first-passage law is log-concave (Conjecture 10.1 of UN; evidence: exact arithmetic at $q = \frac{1}{2}$ on windows of six modal times for $d = 2$, $N \in \{2, \dots, 14, 16, 18, 20\}$, and $d = 3$, $N \leq 8$). (b) In continuous time, for every pair, g' has at most three sign changes. (c) For $d \in \{1, 2, 3\}$, $q \in \{\frac{4}{5}, \frac{1}{2}\}$ and every N , with t^* the largest maximiser, every occupation probability is already decreasing at t^* : $v_{t^*+1} < v_{t^*}$ (Conjecture 47 of CE; a theorem on the certified ranges, Computer-assisted theorem 6.13).

6.6 Certified modes in two and three dimensions

Computer-assisted theorem 6.13 (exact modes, $d = 2, 3$; [CE, Thms 12, 39, Cors 18, 42]). For the CC pair, $q = \frac{4}{5}$: $d = 2$, $2 \leq N \leq 301$ and $N \in \{350, 401, 450, 500, 600\}$; $d = 3$, $2 \leq N \leq 120$ and $N \in \{130, 140, 150, 160\}$; $q = \frac{1}{2}$: $d = 2$, $2 \leq N \leq 200$; $d = 3$, $2 \leq N \leq 80$. In all these cases $f(t) = 0$ for $t < d(N-1)$, the PMF is strictly unimodal with the certified integer mode (e.g. $d = 2$, $N = 600$: $t^* = 872259$; $d = 3$, $N = 160$: $t^* = 170871$, both at $q = \frac{4}{5}$), and every occupation probability is already decreasing at the mode ($v_{t^*+1} < v_{t^*}$, tail time $T = t^*$). Moreover $\frac{4}{5}t^*(\frac{4}{5}) - \frac{1}{2}t^*(\frac{1}{2}) \in [-0.9, 0.3]$ ($d = 2$, $N \leq 200$) and $[-1.2, 0.4]$ ($d = 3$, $N \leq 80$): the discrete mode is a function of qt up to about one time step.

The ranges $d = 2$, $N \leq 160$ ($q = \frac{4}{5}$), $N \leq 80$ ($q = \frac{1}{2}$) and $d = 3$, $N \leq 60$ ($q = \frac{4}{5}$), $N \leq 32$ ($q = \frac{1}{2}$) rest on Python integers only; the rest rests in addition on the C engine (Section 9). These certified integers agree with all 97 corner-to-corner values of the computations of Sections 5–6 of the article that fall in the certified sets.

7 Spectral quantities of the corner-to-corner box

In this section $\mathcal{L}$ is the box generator, CC geometry, $u = \frac{\pi}{2N}$, $m = m(x_0)$ the unit-rate corner-to-corner MFPT, τ the unit-rate mean hitting time of the corner from the uniform start, $\bar{X} = \Lambda_1 \tau$, $X = \Lambda_1 m$, and for activity q all times are divided by q .

7.1 Every dimension

Lemma 7.1 (arrival profile; [SP, Lemma 7.2]).

$$n p_t(x_0, a) = u_1(t/d)^d, \quad u_1(s) = 1 + 2 \sum_{k=1}^{N-1} (-1)^k \cos^2(ku) e^{-(1-\cos \frac{\pi k}{N})s},$$

and $0 \leq 1 - n p_t(x_0, a) \leq \min\{1, W e^{-\Lambda_1 t}\}$. Hence

$$\frac{d(N^2 - 1)}{3} \leq m - \tau = \int_0^\infty (1 - n p_t(x_0, a)) dt \leq \frac{1 + \ln W}{\Lambda_1},$$

with equality on the left iff $d = 1$; in particular $m - \tau \geq 1/\Lambda_1$.

Lemma 7.2 (second-moment functional; [SP, Lemmas 7.4, 7.7]). $W + \binom{d}{2} \cos^4 u \leq \Lambda_1^2 K_2 \leq Z_d(N) := \sum_{0 < |k|_\infty \leq N-1} |k|^{-4} \leq Z_d := \sum_{k \in \mathbb{Z}^d \setminus \{0\}} |k|^{-4}$ ($d \leq 3$), and $\bar{X} \geq \frac{4}{d} N^{d-2} (1 - N^{-d})^2$; for $d \geq 3$, $\bar{X}$ is of exact order N^{d-2} .

Lemma 7.3 (rational enclosure of the MFPT; [CE, Lemma 19]). The vector $m = (m(x))_{x \in \Lambda'}$ of unit-rate mean hitting times is the unique solution of $Km = 2d\mathbf{1}$, $K = 2dI - (A + B)$ on Λ' (A the adjacency matrix, $B = \text{diag}(b)$), and $K^{-1} \geq 0$ entrywise. Hence for every vector z with $R := Kz > 0$ entrywise, $\frac{2d}{\max R} z \leq m \leq \frac{2d}{\min R} z$.

Proof. $K = 2d(I - Q_1)$ and $(I - Q_1)^{-1} = \sum_k Q_1^k \geq 0$ because the spectral radius of Q_1 is < 1 (absorption is certain); $m = \sum_t Q_1^t \mathbf{1}$. Then $K(\frac{2d}{\min R} z - m) = 2d(\frac{R}{\min R} - \mathbf{1}) \geq 0$; apply $K^{-1} \geq 0$. The lower bound is symmetric. $\square$

A floating-point guess z , rounded to integers, gives intervals of relative width below 10^{-28} [CE, Section 5].

Computer-assisted lemma 7.4 (Epstein-type sums; [SP, Lemma 7.6]). With $Z'_d = \sum_{|k|^2 \geq 2} (|k|^2(|k|^2 - 1))^{-1}$ (truncated decimals, every digit certified): $Z_2 = 6.02681203969194012\dots$, $Z_3 = 16.5323159597616696\dots$, $Z'_2 = 3.22591076832683889\dots$, $Z'_3 = 14.7022972380072310\dots$ (theta splitting with Jacobi's transformation, incomplete gamma functions and explicit tail bounds, 300-bit Arb). Every printed string is asserted by the certification script.

Lemma 7.5 (monotone coupling; [SP, Lemma 7.8], [UN, Lemma 5.1]). For the corner target and $x \preceq y$ coordinatewise, $\mathbb{P}_x(T > t) \geq \mathbb{P}_y(T > t)$ for all t , in continuous and discrete time and every q . Hence $m(\cdot)$ and $r_0(\cdot)$ are maximal at the opposite corner.

Proof. Drive two copies by the same proposed moves; the maps $x_i \mapsto \min(x_i + 1, N)$ and $x_i \mapsto \max(x_i - 1, 1)$ are nondecreasing, so the order is preserved, and when the lower copy reaches the maximal site a the upper copy is there too. $\square$

Theorem 7.6 ($r_0 > 1$ and $\sigma_0 m > 1$; [SP, Thm 7.9]). For every $d \geq 1$, $N \geq 2$, $(d, N) \neq (1, 2)$: $r_0 = \max_x r_0(x) > 1$ and $\sigma_0 m = \max_x \sigma_0 m(x) > 1$; quantitatively $\sigma_0 m \geq (\bar{X} + \mu_-)/(\bar{X} + 1)$ with $\mu_- = \frac{2}{3}(N^2 - 1) \sin^2 u \geq 1$.

Proof. By Theorem 4.3(e) the ψ_0 -weighted averages of $r_0(\cdot)$ and $\sigma_0 m(\cdot)$ are 1; by Lemma 7.5 both are maximal at x_0 ; equality would force them to be constant, i.e. every site of Λ' to be a neighbour of a , which happens only for $(d, N) = (1, 2)$. The quantitative bound is Theorem 4.15 with Lemma 7.1. $\square$

Theorem 7.7 (CC, explicit in $\bar{X}$; [SP, Thm 7.10, Cor 7.11]). Let $d \geq 2$, $Z \geq \Lambda_1^2 K_2$ (e.g. $Z = Z_d$ for $d \leq 3$), $\ell_W = 1 + \ln W$, $\bar{X} > 1$, $\beta = \bar{X}/(\bar{X} - 1)$. Then

$$1 - \frac{\beta Z}{\bar{X}^2} \leq \sigma_0 \tau \leq 1 - \frac{W}{\bar{X}^2} + \frac{2W^2}{\bar{X}^4}, \quad \left(1 + \frac{\mu_-}{\bar{X}}\right) \left(1 - \frac{\beta Z}{\bar{X}^2}\right) \leq \sigma_0 m < 1 + \frac{\ell_W}{\bar{X}},$$

always $\tau \leq 1/\sigma_0 \leq \tau + 1/\Lambda_1 \leq m$, $\Lambda_1(1 - \bar{X}^{-1}) < \sigma_1 - \sigma_0 < 2\Lambda_1$, $r_0 > 1$, $r_1 < 0$, and explicit two-sided bounds hold for σ_1 , $\sigma_1 - \sigma_0$, r_0 and r_1 (SP, Thm 7.10(iii)–(vi)). For $d \geq 3$: $0 < 1 - \sigma_0 \tau \leq \beta Z_d(N)/\bar{X}^2$ and $0 < \sigma_0 m - 1 < (1 + \ln 2d)/\bar{X}$, so $\sigma_0 \tau = 1 - O(N^{-2})$ ($d = 3$), $1 - O(N^{-4} \ln N)$ ($d = 4$), $1 - O(N^{-d})$ ($d \geq 5$) with explicit constants.

Proposition 7.8 (exact reductions; [SP, Lemmas 8.1, 8.2, Props 8.3, 8.5]). For $d = 2$, with $x_k = \frac{\pi k}{2N}$, $y_k = 2N \operatorname{arsinh}(\sin x_k)$, $\Phi(x) = \cos^2 x \sqrt{1 + \sin^2 x} / \sin x$:

$$\tau_2 = 4N \sum_{k=1}^{N-1} \Phi(x_k) \coth y_k - \frac{2}{3}(N^2 - 1), \quad m_2 = 4N \sum_{k=1}^{N-1} \Phi(x_k) \left[\coth y_k - \frac{(-1)^k}{\sinh y_k} \right].$$

For $d = 3$, τ_3 and $m_3 - \tau_3$ are double sums of the same kind.

Proof (outline). One of the d summations in $\tau = \sum_{k \neq 0} w_k / \lambda_k$ is done in closed form by the Poisson-kernel identity $\sum_{k=0}^{N-1} \varepsilon'_k \frac{1 + \cos \theta_k}{\cosh \varphi - \cos \theta_k} = N[\coth \frac{\varphi}{2} \coth N\varphi - 1]$ and its alternating analogue, plus cotangent sums. Complete proof: [SP, Section 8]. The formula for m_2 is the single-sum identity (Theorem 4.2 of the article). $\square$

7.2 Two dimensions

Let γ be Euler's constant and (Lemma 9.2 of SP, from DLMF theta and elliptic-integral identities at the lemniscatic point)

$$c_\tau = \frac{8}{\pi} \left(\gamma - \frac{1}{2} + \frac{7}{2} \ln 2 + \frac{1}{2} \ln \pi - 2 \ln \Gamma\left(\frac{1}{4}\right) \right) - 2 = -0.727904612791689 \dots,$$

$$c_{m\tau} = \frac{4 \ln 2}{\pi} = 0.882542400610606 \dots$$

Computer-assisted theorem 7.9 ($d = 2$ with explicit remainder; [SP, Thm 9.1]). *For every $N \geq 2$: $\tau_2 = \frac{8}{\pi} N^2 \ln N + c_\tau N^2 + R_\tau(N)$ with $-2.14 \leq R_\tau(N) \leq 3.53$, and $m_2 - \tau_2 = c_{m\tau} N^2 + R_{m\tau}(N)$ with $-9.06 \leq R_{m\tau}(N) \leq 7.72$. Consequently*

$$m_2 = \frac{8}{\pi} N^2 \ln N + C_2 N^2 + R_m(N), \quad C_2 = c_\tau + c_{m\tau} = 0.154637787818917 \dots, \quad |R_m(N)| \leq 11.3.$$

The proof is analytic except for the numerical constants (a bound on $|\Psi''|$ and four rapidly convergent series), which are certified in ball arithmetic.

Theorem 7.10 ($d = 2$: the constant terms; [SP, Lemma 9.6, Thm 9.8]). [†] *As $N \rightarrow \infty$, $R_\tau(N) \rightarrow c_\tau^{(0)} = \frac{2}{3} + \frac{\pi}{18} - \frac{2\pi}{3} S_5 + \frac{2\pi^2}{3} S_6 = 0.8830352 \dots$ and $R_{m\tau}(N) \rightarrow c_{m\tau}^{(0)} = -\frac{2}{3} - \frac{2\pi}{3} S_7 + \frac{2\pi^2}{3} S_8 = -0.3508943 \dots$, with $S_5 = \sum_k \frac{2k}{e^{2\pi k} - 1}$, $S_6 = \sum_k \frac{k^2}{\sinh^2 \pi k}$, $S_7 = \sum_k \frac{(-1)^{k+1} k}{\sinh \pi k}$, $S_8 = \sum_k \frac{(-1)^{k+1} k^2 \cosh \pi k}{\sinh^2 \pi k}$. Hence $m_2 = \frac{8}{\pi} N^2 \ln N + C_2 N^2 + C_2^{(0)} + o(1)$ with $C_2^{(0)} = 0.5321408 \dots$. The limits are proved analytically (dominated convergence with the N -independent majorants of the proof of Theorem 7.9, and a Riemann-sum limit in Lemma 9.6 of SP); only the decimal values are computer-assisted; no rate is claimed.*

Proof (outline). In the decomposition $R_\tau = \frac{2}{3} - \frac{8N^2}{\pi} \eta_N + 4N e_1(N) + 4N E_c(N)$ of the proof of Theorem 7.9, $\frac{8N^2}{\pi} \eta_N \rightarrow \frac{2}{3\pi}$ (harmonic-number bounds), $4N e_1(N) \rightarrow \frac{2}{3\pi} + \frac{\pi}{18}$ (mean value theorem on each cell and a Riemann sum), and $N E_c(N) = \sum_k a_k(N) + \sum_k b_k(N) - N \cdot (\cdot)$ with summands bounded by N -independent summable majorants and with explicit pointwise limits, so Tannery's theorem gives $-\frac{\pi}{6} S_5 + \frac{\pi^2}{6} S_6$; the same for $R_{m\tau}$. Complete proof: [SP, Section 9]. $\square$

Classical Eisenstein-series evaluations at the lemniscatic point give the closed form $C_2^{(0)} = \Gamma(\frac{1}{4})^8 / (576\pi^4)$, i.e. $C_2^{(0)} = 0.5321408832 \dots$; this closed form is quoted, not proved, and not used; it agrees with the certified value of the series to more than 50 digits (Remark 9.9 of SP). The monotonicity of $R_\tau(N)$, $R_{m\tau}(N)$ in N , the ranges $[0.8512, 0.8831]$ and $[-0.5302, -0.3508]$ for all N , and the rate N^{-2} are numerical observations [SP, Obs 9.7].

The certificate note encloses the same constant and the remainder on a finite range, by a different route (rational enclosures of the MFPT from the linear system).

Computer-assisted theorem 7.11 (value of C_2 ; [CE, Thm 25]). *With the lemniscate constant $\varpi = \Gamma(\frac{1}{4})^2 / (2\sqrt{2\pi})$,*

$$\begin{aligned} C_2 &= \frac{8}{\pi} \left[\gamma + \ln \frac{4\sqrt{2}}{\varpi} - \frac{1}{2} - \frac{\pi}{4} \right] \\ &= \frac{8}{\pi} \left[\gamma + 4 \ln 2 + \frac{1}{2} \ln \pi - 2 \ln \Gamma\left(\frac{1}{4}\right) \right] - 2 - \frac{4}{\pi}, \end{aligned}$$

and $|C_2 - c_o| < 10^{-50}$ with

$$c_o = 0.15463778781891726249073383080028304919172661983273.$$

Remark 7.12 (identification of C_2). [‡] The two closed forms of C_2 coincide: $c_\tau + c_{m\tau} = \frac{8}{\pi} (\gamma - \frac{1}{2} + 4 \ln 2 + \frac{1}{2} \ln \pi - 2 \ln \Gamma(\frac{1}{4})) - 2$, which is the second expression of Computer-assisted theorem 7.11. Hence, by Computer-assisted theorem 7.9, C_2 is the coefficient of N^2 in m_2 , with an explicit $O(1)$ remainder valid for every N (the identity of the two closed forms is elementary; the remainder bound 11.3 is computer-assisted); this closes the gap [CE, G3], where the identification rested on a sketched remainder estimate.

Computer-assisted theorem 7.13 (two-term law on a finite range; [CE, Thm 26, Prop 27]). *Let $r_N := m_2 - \frac{8}{\pi} N^2 \ln N - C_2 N^2$. For $2 \leq N \leq 301$, $0.3211 < r_N < 0.53213$ and r_N is strictly increasing (from rational enclosures of m_2 , Lemma 7.3). Moreover[‡] $r_N < 0.5321405 < C_0 := \varpi^4 / (9\pi^2) = 0.53214088 \dots$ for $301 < N \leq 1500$, and r_N is strictly increasing on the whole range $2 \leq N \leq 1500$.*

Proof. The first part is [CE, Thm 26]. In [CE, Prop 27] the second part is proved under the hypothesis $m_2 = 4N \sum_{k=1}^{N-1} c_k^2 \frac{\sqrt{1+s_k^2}}{s_k} g_k$ ($s_k = \sin \frac{\pi k}{2N}$, $c_k = \cos \frac{\pi k}{2N}$, $g_k = \coth(N \operatorname{arsinh} s_k)$ for odd k and $\tanh(N \operatorname{arsinh} s_k)$ for even k), including the junction $r_{301} < r_{302}$ (an inequality between the upper bound 0.532129078393 of r_{301} and the lower bound 0.532129156439 of r_{302} , both rounded outwards[†]). This hypothesis is the m_2 -formula of Proposition 7.8: since $\coth y + \frac{1}{\sinh y} = \coth \frac{y}{2}$ and $\coth y - \frac{1}{\sinh y} = \tanh \frac{y}{2}$, and $\frac{y_k}{2} = N \operatorname{arsinh}(\sin x_k)$, the bracket $\coth y_k - (-1)^k / \sinh y_k$ is g_k , and $\Phi(x_k) = c_k^2 \sqrt{1+s_k^2}/s_k$. So the hypothesis holds, and the statement is unconditional. $\square$

This is consistent with Theorem 7.10: $r_N = R_m(N)$ increases towards its limit $C_2^{(0)} = 0.5321408\dots$ on the computed range. The certified values of C_0 and of the series $C_2^{(0)}$ agree to more than 50 digits; their equality is the quoted Eisenstein-series closed form mentioned above, which is not proved here.

7.3 Three dimensions

Computer-assisted theorem 7.14 ($d = 3$; [SP, Thm 10.1]). (a)

$$C_3 = \frac{3}{\pi^3} \int_{[0,\pi]^3} \frac{\prod_i (1 + \cos \theta_i)}{3 - \sum_i \cos \theta_i} d\theta = 3 \int_0^\infty [e^{-s}(I_0(s) + I_1(s))]^3 ds = \sum_{y \in \{0,1\}^3} u(y)$$

with u the Green's function of the simple random walk on $\mathbb{Z}^3$, and $4.320460169373 < C_3 < 4.320460169375$.
(b) For every $N \geq 2$:

$$C_3 N^3 - 8.51 N^2 - 2.54 \leq \tau_3 \leq C_3 N^3 - 0.53 N^2 + 3.90, \quad N^2 - 1 \leq m_3 - \tau_3 \leq \frac{3}{2}(1 + \ln 6) N^2.$$

(c) For every $N \geq 2$: $\tau_3 = C_3 N^3 + c_{\tau 3} N^2 + R_3(N)$, $-17.4 \leq R_3 \leq 19.4$, and $m_3 - \tau_3 = c_{m\tau 3} N^2 + R'_3(N)$, $-82.4 \leq R'_3 \leq 80.4$, with $c_{\tau 3} = -5.418839026578\dots$, $c_{m\tau 3} = 1.531584884077\dots$. (d) $R_3(N) \rightarrow 1.7573985\dots$ and $R'_3(N) \rightarrow -0.2989172\dots$; hence $m_3 = C_3 N^3 + C'_3 N^2 + C''_3 + o(1)$ with $C'_3 = c_{\tau 3} + c_{m\tau 3} = -3.887254142500\dots$ and $C''_3 = 1.4584812\dots$ (limits proved analytically, values computer-assisted).

This proves the values $C_3 = 4.32046\dots$, $C'_3 = -3.887\dots$, which are K_3 , K'_3 of Theorem 4.8 of the article (numerical estimates in its Supplementary Section S5.3), with an explicit $O(1)$ remainder for every N ; it closes the gap [CE, G4]. The proved windows for R_3 , R'_3 are wide compared with the observed ranges $[1.61, 1.76]$, $[-0.63, -0.29]$ [SP, Rem 10.12].

7.4 Consequences for the poles

Computer-assisted corollary 7.15 ($\bar{X}$ and $\Lambda_1(m - \tau)$ explicitly; [SP, Cors 11.1, 11.2]). For every $N \geq 2$, with the exact constants $\frac{\pi^2}{4} c_\tau = -1.7960326\dots$, $\frac{\pi^2}{6} C_3 = 7.1068721\dots$, $\frac{\pi^2}{6} c_{\tau 3} = -8.9136329\dots$, $\frac{\pi^2}{6} c_{m\tau 3} = 2.5193561\dots$:

$$d = 2: \left| \bar{X} - 2\pi \ln N - \frac{\pi^2}{4} c_\tau \right| \leq \frac{8.8 + 5.2 \ln N}{N^2}, \quad |\Lambda_1(m_2 - \tau_2) - \pi \ln 2| \leq \frac{24.2}{N^2};$$

$$d = 3: -\frac{5.85}{N} - \frac{21.3}{N^2} \leq \bar{X} - \frac{\pi^2}{6} (C_3 N + c_{\tau 3}) \leq \frac{31.92}{N^2}, \quad |\Lambda_1(m_3 - \tau_3) - \frac{\pi^2}{6} c_{m\tau 3}| \leq \frac{137.7}{N^2}.$$

All bounds of Theorem 7.7 then become explicit functions of N (for $N \geq 3$; void for $N = 2$).

Refuted statement 7.16 (truncated constants; [SP, Prop 11.5]). [†] The constants of Corollary 7.15 may not be rounded: with the truncated decimals 1.796034 and 7.1069 the two bounds become false: $|\bar{X} - (2\pi \ln N - 1.796034)| \leq (8.8 + 5.2 \ln N)/N^2$ fails for every $N \geq 8872$ (and already at $N = 8540, 10^4$, by direct evaluation; it holds at $N = 8539$), and $7.1069N - 14.0 - 5.85/N - 4.18/N^2 \leq \bar{X}$ fails for every $N \geq 182\,420$. An $O(N^{-2})$ error term cannot absorb a constant that is wrong in the sixth decimal.

Theorem 7.17 (first-order pole structure, $d = 2, 3$; [SP, Thm 11.3]). As $N \rightarrow \infty$: $\sigma_0 \tau = 1 - Z_d \bar{X}^{-2} + o(\bar{X}^{-2})$; $\Lambda_1(m - \tau) \rightarrow \mu_\infty$ ($\mu_\infty = \pi \ln 2 = 2.17758\dots$ for $d = 2$, $\frac{\pi^2}{6} c_{m\tau 3} = 2.51935\dots$ for $d = 3$), so $X = \bar{X} + \mu_\infty + o(1)$ and $\sigma_0 m = 1 + \mu_\infty / \bar{X} + o(\bar{X}^{-1})$; $\sigma_1 / \Lambda_1 = 1 + \delta$ with $W / \delta = \bar{X} - (2d + 1 - Z'_d) + o(1)$; $(\sigma_1 - \sigma_0) / \Lambda_1 = 1 + (2d - 1) / \bar{X} + O(\bar{X}^{-2})$; $r_0 = 1 + \mu_\infty / \bar{X} + o(\bar{X}^{-1})$, $r_1 = -\frac{2d}{\bar{X}} (1 + O(\bar{X}^{-1}))$, $r_1 \sigma_1 / (r_0 \sigma_0) \rightarrow -2d$.

These are the relations “ $\nu_0 = \text{MFPT}^{-1}(1 + O(1/X))$ ”, $\nu_1 = \mu_1(1 + W/X + \dots)$, $a_1 \approx -A\nu_0$ ” of the formal two-pole argument, proved here; the rate $\mu_1(X + W - 1)/X$ that underlies the closed form is (d) of [SP, Thm 11.3]. They do not by themselves locate the mode [SP, G1]; the mode theorems of Section 8 are proved by a different route.

8 The mode in two and three dimensions and the closed form

Throughout this section the geometry is CC, $d \in \{2, 3\}$, $x = \frac{\pi}{2N}$, $\mu = \mu_1 = \frac{2q}{d} \sin^2 x$, and times are scaled, $\tau = \mu t$; $\tau^* = \mu t^*$ and $\tau_{\text{cf}} = X \ln(2dX)/(X + 2d - 1)$. The results are those of AS, except the finite-range statements of the last subsection; they use Section 4 and, in discrete time with $q \leq \frac{1}{2}$, Theorem 6.1, but none of the estimates of Section 7. The finite-range statements (Computer-assisted theorem 8.14, Computer-assisted corollary 8.15) use in addition Computer-assisted theorem 6.13 and Lemma 7.3.

8.1 Decomposition and unimodality

Let $\varepsilon_k = \sin^2(kx)/\sin^2 x$, $\gamma_0 = 1$, $\gamma_k = 2 \cos^2(kx)$, and for $k \in \{0, \dots, N-1\}^d \setminus \{0\}$, $\mu_k = \sum_i \varepsilon_{k_i}$, $w_k = \prod_i \gamma_{k_i}$, $c_k = (-1)^{|k|} w_k$. Let $v(\tau) = \sum_k (-1)^k \gamma_k e^{-\varepsilon_k \tau}$ (one coordinate), $u = v^d$, $\varphi = u'$.

Proposition 8.1 (decomposition; [AS, Lemmas 1.2–1.4, 1.7, Prop 1.5, Rem 1.6]). *v is the distribution function of a sum of $N - 1$ independent exponential variables with rates $\varepsilon_1, \dots, \varepsilon_{N-1}$, and $\mu T \stackrel{d}{=} U + \mu T_\pi$, where U has distribution function $u = v^d$, T_π is the hitting time of a from the uniform start, and the two are independent. In continuous time μT has the density*

$$g(\tau) = \pi_a \varphi(\tau) + \int_0^\tau \varphi(\tau - s) h_{\text{ac}}(s) ds = \sum_{j=0}^{L-1} \rho_j \hat{\varphi}(-\nu_j) e^{-\nu_j \tau}, \quad \pi_a = N^{-d},$$

where the poles $0 < \nu_0 < 1 < \nu_1 < 2 < \dots$ interlace the distinct levels of μ_k and $\rho_j > 0$; in discrete time $\mathbb{P}(T = t) = \mu \sum_j \rho_j \hat{\varphi}(-\nu_j) (1 - \mu \nu_j)^{t-1}$. Moreover $X = X_s + \mathbb{E}U$ with $X_s = \sum_k w_k / \mu_k = \bar{X}$.

Theorem 8.2 (continuous time: strict unimodality with a monotone ratio; [AS, Lemma 1.10, Thm 1.11]). *For $d \geq 1, N \geq 2$, $(d, N) \neq (1, 2)$: v' , v and φ are positive and log-concave on $(0, \infty)$, $B = g'/\varphi$ is nonincreasing, and g' has exactly one zero $\tau^* \in (0, \infty)$; $g' > 0$ on $(0, \tau^*)$ and $g' < 0$ on (τ^*, ∞) . (For $(d, N) = (1, 2)$, μT is exponential and there is no interior mode.) The extension from $d \geq 2, N \geq 3$ to $d = 1$ and to $N = 2$ was proved after the internal checks[†]; unimodality on that extended range is in any case Theorem 6.1(i) and Theorem 5.2.*

Proof (outline). Convolutions of log-concave densities on $[0, \infty)$ with exponential kernels, and their primitives, are log-concave; $g' = \pi_a \varphi' + \varphi h_{\text{ac}}(0) - \int_0^\tau \varphi(\tau - s) \eta(s) ds$ with $\eta = -h'_{\text{ac}} > 0$; divide by φ and use that $\varphi(\tau - s)/\varphi(\tau)$ is nondecreasing in τ . $\varphi(0) = 0$ places the maximum in the interior. Complete proof: [AS, Section 1.6]. $\square$

8.2 An exact identity and the window theorem

For $d \in \{2, 3\}$, $N \geq 3$ ($N \geq 4$ if $d = 3$) the levels below $\varepsilon_2 = 4 \cos^2 x \geq 3$ are 1 (weight $W = dA$, $A = 2 \cos^2 x$), 2, and 3 if $d = 3$. Let $\theta = \nu_1 - 1$, $a_1 = W \rho_1 / \theta$, $a_0 = \rho_0 \nu_0 \hat{\varphi}(-\nu_0)$, $\Lambda = a_1 / a_0$, $\hat{\nu} = \nu_1 - \nu_0$.

Theorem 8.3 (exact identity and window theorem; [AS, Prop 2.1, Lemma 2.2, Thm 2.3]).

$$g'(\tau) = a_1 e^{-\nu_1 \tau} \left[1 + \frac{\theta}{W} \kappa(\tau) \right] - a_0 e^{-\nu_0 \tau} [1 + E_-(\tau)] + P_{\text{rest}}(\tau),$$

where every correction term $\kappa, E_-, P_{\text{rest}}$ has a definite sign and explicit two-sided exponential bounds; hence $\Phi_- \leq g' \leq \Phi_+$ with explicit functions $\Phi_\pm$, and by Theorem 8.2, $\Phi_-(\tau_-) > 0 \Rightarrow \tau_- < \tau^*$ and $\Phi_+(\tau_+) < 0 \Rightarrow \tau^* < \tau_+$. Dropping all corrections, g' vanishes at $t_c = \ln \Lambda / \hat{\nu}$; this is the rigorous form of the two-pole heuristic.

Lemma 8.4 (pole data; [AS, Lemmas 3.1–3.5]). *With $y = 1/X_s$: $\nu_0, \rho_0, \hat{\varphi}(-\nu_0), \nu_1, \rho_1$ and the near poles are bounded above and below by explicit functions of y and of the lattice constant $T_1 = \sum_{\mu_k \geq \varepsilon_2} w_k / (\mu_k(\mu_k - 1))$; in particular $\Lambda = W X_s (1 + O(y))$ and $\hat{\nu} = 1 + (W - 1)y + O(y^2)$, so $t_c = \ln(W X_s) (1 - (W - 1)y) + O(y)$.*

Computer-assisted proposition 8.5 (lattice sums; [AS, Props 4.5, 4.6, Cor 4.7]). *For $N \geq 10$: $2\pi \ln N - 2.32 \leq X_s \leq 2\pi \ln N - 1.60$ ($d = 2$); $c_3 N - 3\pi - \frac{5.85}{N} \leq X_s \leq c_3 N - 8.38$ ($d = 3$); $2\pi \ln N - 0.24 \leq X \leq 2\pi \ln N + 0.60$ ($d = 2$); $c_3 N - 7.63 \leq X \leq c_3 N - 5.85$ ($d = 3$); here $c_3 = \frac{\pi^2}{2} \int_0^\infty (e^{-z}(I_0 + I_1)(z))^3 dz \in [7.106860, 7.106880]$, and $2.08 \leq \mathbb{E}U \leq 2.20$ ($d = 2$), $2.39 \leq \mathbb{E}U \leq 2.53$ ($d = 3$).*

Remark 8.6 (consistency with Section 7). $X_s = \bar{X}$ and $\mathbb{E}U = \Lambda_1(m - \tau)$, and $c_3 = \frac{\pi^2}{6} C_3$ with C_3 of Computer-assisted theorem 7.14: indeed $\frac{\pi^2}{6} C_3 = 7.1068721 \dots \in [7.106860, 7.106880]$. The bounds of Computer-assisted proposition 8.5 are consistent with the sharper Computer-assisted corollary 7.15 ($\bar{X} = 2\pi \ln N - 1.7960326 \dots + O(N^{-2} \ln N)$ in $d = 2$; $\Lambda_1(m - \tau) \rightarrow \pi \ln 2 = 2.17758 \dots$, resp. $2.51935 \dots$, inside $[2.08, 2.20]$, resp. $[2.39, 2.53]$). The certificates of this section do not use Section 7; using its sharper bounds would narrow the constants below but would require a new certificate run, which has not been done.

8.3 The closed form in continuous time

Computer-assisted theorem 8.7 (accuracy of the closed form, continuous time; [AS, Thm 5.1]). *For every $N \geq N_0$, $-K_d^-(N_0) < X(\tau^* - \tau_{\text{cf}}) < K_d^+(N_0)$, with*

$d = 2: N_0$	12	15	20	30	50	100	200	1000	10^6
$K_{2,D}^-$	7.653	5.669	4.242	3.023	2.058	1.320	0.767	-0.116	-1.382
$K_{2,D}^+$	10.280	8.774	7.486	6.348	5.449	4.839	4.641	4.641	4.641
$d = 3: N_0$	10	15	20	30	50	100	200	1000	10^6
$K_{3,D}^-$	3.120	1.995	1.564	1.214	1.160	1.068	0.995	0.862	0.828
$K_{3,D}^+$	7.120	4.659	3.663	2.786	2.149	1.740	1.598	1.465	1.427

(a negative K^- means $\tau^* > \tau_{\text{cf}}$). Equivalently $|t^* - t_{\text{cf}}| \leq K_d(N_0)/(X\mu)$ and $|t^* - t_{\text{cf}}|/t_{\text{cf}} \leq K_d(N_0)(1 + \frac{2d-1}{X})/(X \ln(2dX))$, $K_d = \max(K_d^-, K_d^+)$: the relative error of the closed form is $O(1/(\ln N \ln \ln N))$ in $d = 2$ and $O(1/(N \ln N))$ in $d = 3$.

Proof (outline). Lemma 8.4 and Computer-assisted proposition 8.5 turn the sign conditions of Theorem 8.3 at $\tau_{\text{cf}} \pm c_{\pm}/X$ into finitely many inequalities between elementary expressions over parameter boxes. The values of $N \geq 10$ are covered by 220 ($d = 2$) resp. 389 ($d = 3$) blocks, singletons for small N , geometric blocks up to 10^{60} ($d = 2$), 10^9 ($d = 3$), and one unbounded block each; for each block the inequalities are verified in Arb ball arithmetic (code/msc_rigorous_asymptotics/52_cert_theoremB.py with certlib.py). The table entry is the maximum over all blocks that meet $[N_0, \infty)$. Complete proof: [AS, Section 5]. $\square$

Computer-assisted theorem 8.8 (asymptotic laws, continuous time; [AS, Thms 6.1–6.3, Cor 6.4]). *For every $N \geq 20$ and every rate q :*

$$d = 2: \quad t^* = \frac{4N^2}{\pi^2 q} [\ln \ln N + \ln(8\pi) + R_N](1 + \rho_N), \quad -\frac{4.6 + 3 \ln(8\pi \ln N + 2.4)}{2\pi \ln N - 0.24} \leq R_N < 0,$$

$$d = 3: \quad t^* = \frac{6N^2}{\pi^2 q} [\ln N + \ln(6c_3) + R_N](1 + \rho_N), \quad -\frac{9.2 + 5 \ln(6c_3 N)}{c_3 N - 7.63} \leq R_N < 0,$$

with $0 \leq \rho_N \leq 0.83/N^2$ and $\ln(6c_3) \in [3.752820, 3.752823]$. Two-sided bounds: $\frac{4}{\pi^2 q} N^2 (\ln \ln N + 2.27) \leq t^* \leq \frac{4}{\pi^2 q} N^2 (\ln \ln N + 3.2242)(1 + \frac{0.83}{N^2})$ ($d = 2$) and $\frac{6}{\pi^2 q} N^2 (\ln N + 3.43) \leq t^* \leq \frac{6}{\pi^2 q} N^2 (\ln N + 3.7529)(1 + \frac{0.83}{N^2})$ ($d = 3$). Moreover $t^*/\mathbb{E}T = \ln(2dX)/(X + 2d - 1) + \vartheta_N/X^2$ with ϑ_N bounded as in Computer-assisted theorem 8.7; so $t^*/\mathbb{E}T \rightarrow 0$, like $\frac{\ln \ln N}{2\pi \ln N}$ ($d = 2$) and $\frac{\ln N}{c_3 N}$ ($d = 3$).

These are the laws “ $t^* \sim \frac{4}{\pi^2 q} N^2 \ln \ln N$ ” and “ $\frac{6}{\pi^2 q} N^2 (\ln N + 3.753)$ ” of the computations of Sections 5–6 of the article, with proven remainders. In $d = 2$ the leading correction $\ln(8\pi)/\ln \ln N$ decays extremely slowly; the two-term form is the practically relevant statement.

8.4 Discrete time

Computer-assisted theorem 8.9 (discrete time, $q \leq \frac{1}{2}$; [AS, Prop 7.1, Lemmas 7.2, 7.3, 7.6, Thms 7.7, 7.8, Cor 7.9]). *Let $q \in (0, \frac{1}{2}]$. For every $N \geq N_0$ every maximiser of the PMF satisfies*

$$t_{\text{cf}} - \frac{K_{d,D}^-(N_0)}{X\mu} < t^* < t_{\text{cf}} + \frac{K_{d,D}^+(N_0)}{X\mu} + 2,$$

uniformly in q , with

$d = 2: N_0$	12	15	20	30	50	100	200	1000
$K_{2,D}^-$	8.546	6.119	4.479	3.131	2.099	1.332	0.770	-0.116
$K_{2,D}^+$	10.392	8.859	7.542	6.377	5.462	4.842	4.641	4.641
$d = 3: N_0$	10	15	20	30	50	100	200	1000
$K_{3,D}^-$	5.293	3.442	2.661	1.962	1.527	1.318	1.144	0.898
$K_{3,D}^+$	7.513	4.834	3.761	2.829	2.165	1.744	1.599	1.465

Consequently, for $N \geq 20$, the laws of Computer-assisted theorem 8.8 hold for the discrete walk with the lower numerators 4.72 ($d = 2$) and 10.30 ($d = 3$) and with “+2” added to the upper bounds.

Proof (outline). The discrete analogue of Theorem 8.3 (Prop. 7.1, Lemma 7.6, Thm 7.7 of AS, valid for $q \leq \frac{1}{2}$, where all $1 - \mu\nu_j \in (0, 1)$) gives certified signs of the increment $(Dg)(t) = (g(t+1) - g(t))/\mu$ at $t_{\pm}$, $\mathbb{P}(T = t) = \mu g(t-1)$ (script 72, the blocks of the continuous case; the block $d = 2$, $N = 10$ fails and is excluded). To pass from these signs to the location of the maximisers, unimodality of $t \mapsto \mathbb{P}(T = t)$ is used:

if $(Dg)(t) < 0$ every maximiser is $\leq t + 1$, and if $(Dg)(t) > 0$ every maximiser is $\geq t + 2$ (for a unimodal sequence, a strict decrease after $t + 1$ excludes later maximisers and a strict increase excludes earlier ones). In AS unimodality is taken from [AS, Thm 7.4], which rests on the cited closure of ultra-log-concave sequences under convolution (Section 6); here[†] it is taken from Theorem 6.1(ii), proved by hand, which applies since $q \leq \frac{1}{2} < q_N^\square$ and covers every d and N . Hence the theorem does not depend on that citation. $\square$

For $q > \frac{1}{2}$ some multipliers are negative and the PMF need not be unimodal. Let $N_1(q) := \min\{N \geq 4 : \sin \frac{\pi}{N} \leq \frac{1-q}{q}\}$ ($N_1(q) = 4$ for $q \leq \frac{1}{2}$, $N_1(q) \approx \pi q/(1-q)$ as $q \rightarrow 1$); the hypothesis (H) of AS is $q < 1$ and $N \geq N_1(q)$.

Theorem 8.10 (localisation without unimodality; [AS, Lemmas 7.10–7.12, Thm 7.13]). *Assume (H). If integers $t_E \leq t_B \leq t_- < t_+$ satisfy four explicit inequalities (A)–(D) (an upper sign condition at t_+ , a lower sign condition on $[t_B, t_-]$, a monotonicity condition on $[t_E, t_B]$, and a value comparison bounding $\mathbb{P}(T = t)$ for $t \leq t_E$ by $\mathbb{P}(T = t_- + 1)$), then every maximiser satisfies $t_- + 2 \leq t^* \leq t_+ + 1$. The proof uses a pairing lemma for the one-dimensional factors (the generating functions $\prod_{k \geq j} (1 - \beta_k z)^{-1}$ have nonnegative coefficients under (H)) and a bound for the poles with negative multiplier.*

Computer-assisted theorem 8.11 (discrete time, $q \leq q_{\max} < 1$; [AS, Thm 7.14, Cor 7.15]). *Let $q_{\max} \in \{0.8, 0.9, 0.95, 0.99\}$ and $q \in (0, q_{\max}]$. For every $N \geq N_0$ every maximiser satisfies $t_{\text{cf}} - K^-(X\mu) < t^* < t_{\text{cf}} + K^+(X\mu) + 2$, with, for example, $q_{\max} = 0.8$: $(N_0; K^-, K^+) = (37; 2.850, 6.613)$, $(100; 1.448, 5.560)$, $(1000; -0.016, 5.135)$ for $d = 2$ and $(13; 5.189, 6.634)$, $(50; 1.851, 2.973)$, $(1000; 1.021, 2.141)$ for $d = 3$; $q_{\max} = 0.99$: $(312; 0.550, 5.135)$ ($d = 2$), $(312; 1.236, 2.231)$ ($d = 3$) (full tables in [AS, Thm 7.14]). For $q \leq 0.99$ and $N \geq N_0(q)$ ($N_0 = 100$ for $q \leq 0.95$, 312 for $0.95 < q \leq 0.99$) the laws of Computer-assisted theorem 8.8 hold for the discrete walk with lower numerators 1.72 ($d = 2$) and 9.28 ($d = 3$) and “+2” in the upper bounds.*

For the value $q = \frac{4}{5}$ this gives $|t^* - t_{\text{cf}}| < \frac{6.613}{X\mu} + 2$ for all $N \geq 37$ ($d = 2$) and $< \frac{6.634}{X\mu} + 2$ for all $N \geq 13$ ($d = 3$).

Lemma 8.12 (constants uniform in q ; [AS, Lemma 7.16]). [†] *Under (H) and $N \geq 10$, with $D_1(x) = 2 \sin 2x - \frac{2}{d} \sin^2 x$, $D_2(x) = 2 \sin 2x - \frac{4}{d} \sin^2 x$, $D_{\varepsilon_2}(x) = \sin 2x(2 - \frac{2}{d} \sin 2x)$: $2(1-q) - a\mu \geq qD_a(x) > 0$ for $a \in \{1, 2, \varepsilon_2\}$ and $q/(1-q) \leq 1/\sin 2x$; the functions D_a increase on $(0, \frac{\pi}{20}]$, and $N \mapsto ND_a(\frac{\pi}{2N})$, $N \mapsto N \sin \frac{\pi}{N}$ increase for $N \geq 10$. Hence the q -dependent coefficients of the quantities n_q, ω_q, ω'_q and $q\pi_a/(1-q)$ of Theorem 8.10 are bounded by functions of N alone (the remaining dependence on q , through $\mu \leq \frac{2}{d} \sin^2 x$ in the factors $(1 - a\mu)^t$ and $1/(1 - \mu)$, is handled as in Computer-assisted theorem 8.11 with $q_{\max}$ replaced by 1).*

Proof. By (H), $2(1-q) \geq 2q \sin 2x$, and $a\mu = q \frac{2a}{d} \sin^2 x$, with $\frac{2\varepsilon_2}{d} \sin^2 x = \frac{2}{d} \sin^2 2x$ for $a = \varepsilon_2 = 4 \cos^2 x$; so $2(1-q) - a\mu \geq qD_a(x)$, and $D_a > 0$ because $4 \sin x \cos x > \frac{2a}{d} \sin^2 x$ for $a \leq 2$, $x \leq \frac{\pi}{20}$, and $\sin 2x < 1 \leq \frac{d}{2}$. For $a \in \{1, 2\}$, $D'_a(x) = 4 \cos 2x - \frac{2a}{d} \sin 2x \geq 4 \cos \frac{\pi}{10} - 2 \sin \frac{\pi}{10} > 0$; $D_{\varepsilon_2} = h(\sin 2x)$ with $h(s) = 2s - \frac{2}{d} s^2$ increasing for $s < \frac{d}{2}$. $N \sin \frac{\pi}{N} = \pi \frac{\sin u}{u} \big|_{u=\pi/N}$ increases with N and $N \sin^2 \frac{\pi}{2N} = \frac{\pi}{2} \sin v \frac{\sin v}{v} \big|_{v=\pi/2N}$ decreases, which gives the monotonicity of $ND_a(\frac{\pi}{2N})$ for $a \in \{1, 2\}$; and $ND_{\varepsilon_2}(\frac{\pi}{2N}) = N \sin \frac{\pi}{N} (2 - \frac{2}{d} \sin \frac{\pi}{N})$ is a product of two positive increasing factors. Inserting these bounds into the definitions of the constants gives the last claim [AS, Section 7.6]. $\square$

Computer-assisted theorem 8.13 (discrete time, every $q < 1$; [AS, Thm 7.17, Cor 7.18]). [†] *Let $N_* = 59$ ($d = 2$), $N_* = 12$ ($d = 3$). For every $q \in (0, 1)$ and every $N \geq \max(N_0, N_1(q))$ with $N_0 \geq N_*$, every maximiser satisfies $t_{\text{cf}} - K_{d,u}^-(X\mu) < t^* < t_{\text{cf}} + K_{d,u}^+(X\mu) + 2$ with constants independent of q : $(59; 2.082, 6.061)$, $(100; 1.508, 5.594)$, $(1000; -0.009, 5.135)$ for $d = 2$; $(12; 6.379, 7.183)$, $(100; 1.675, 2.495)$, $(1000; 1.036, 2.142)$ for $d = 3$. Consequently the laws of Computer-assisted theorem 8.8 hold for the discrete walk for every $q < 1$ and $N \geq \max(N_*, N_1(q))$, with lower numerators 2.33 ($d = 2$) and 14.01 ($d = 3$) (1.75 and 9.31 for $N \geq \max(100, N_1(q))$).*

Proof (outline). Theorem 8.10 remains true with the q -dependent constants replaced by the q -free majorants of Lemma 8.12; the four conditions are then certified block by block in ball arithmetic with $q_{\max}$ replaced by 1 (code/msc_rigorous_asymptotics/78_cert_allq.py; 171 resp. 387 blocks pass). The certificate was written and run after the internal checks and has not been re-checked by separately written code. $\square$

8.5 Certified accuracy on finite ranges, and a combination

Computer-assisted theorem 8.14 (accuracy of the closed form on finite ranges; [CE, Thms 21, 43, Rems 22, 44]). *With the certified integer modes of Computer-assisted theorem 6.13 and the rational MFPT enclosures of Lemma 7.3: (a) $d = 2$: $|t^* - t_{\text{cf}}| \leq 0.00970 t^*$ for $q = \frac{4}{5}$, $10 \leq N \leq 301$ and $N \in \{350, 401, 450, 500, 600\}$ (worst case $N = 137$); $\leq 0.00971 t^*$ for $q = \frac{1}{2}$, $10 \leq N \leq 200$ (worst $N = 133$). (b) $d = 3$: $|t^* - t_{\text{cf}}| \leq$*

$0.00275t^*$ for $q = \frac{4}{5}$, $10 \leq N \leq 120$ and $N \in \{130, 140, 150, 160\}$; $\leq 0.00157t^*$ for $q = \frac{1}{2}$, $10 \leq N \leq 80$. In $d = 2$ the relative error $(t_{\text{cf}} - t^*)/t^*$ is positive for $N \leq 13$ and negative for every $N \geq 14$ of the ranges, with a flat maximum of its modulus near $N = 137$; in $d = 3$ it is positive for every $N \geq 3$ of the ranges at $q = \frac{4}{5}$ and falls to 3.17888×10^{-5} (rounded) at $N = 160$, not monotonically. For $2 \leq N \leq 9$ the closed form is poorer (up to 32% at $N = 2$, $d = 2$).

Computer-assisted corollary 8.15 (the closed form to within 0.3% in three dimensions, for every $N \geq 10$).
[‡] For $d = 3$, $q \in \{\frac{4}{5}, \frac{1}{2}\}$ and every $N \geq 10$: $|t^* - t_{\text{cf}}| \leq 0.0030t^*$. For $d = 2$: $|t^* - t_{\text{cf}}| \leq 0.0100t^*$ for $q = \frac{4}{5}$ when $10 \leq N \leq 301$, $N \in \{350, 401, 450, 500, 600\}$ or $N \geq 1\,761\,450$, and for $q = \frac{1}{2}$ when $10 \leq N \leq 200$ or $N \geq 554\,846$.

Proof. For a window $-K^-(X\mu) < t^* - t_{\text{cf}} < K^+(X\mu) + 2$ one has $t^* > (\tau_{\text{cf}} - \max(K^-, 0)/X)/\mu$ and $|t^* - t_{\text{cf}}| < \max(K^+/X + 2\mu, K^-/X)/\mu$, so

$$\frac{|t^* - t_{\text{cf}}|}{t^*} < B(N) := \frac{\max(K^+/X + 2\mu, K^-/X)}{\tau_{\text{cf}}(X) - \max(K^-, 0)/X}.$$

Insert $X \geq X_{10}(N)$ from Computer-assisted proposition 8.5 ($N \geq 10$), $\mu \leq \frac{2q}{d}(\frac{\pi}{2N})^2$, and use that $\tau_{\text{cf}}(X) = X \ln(2dX)/(X + 2d - 1)$ is increasing in X (its derivative is $((2d - 1) \ln(2dX) + X + 2d - 1)/(X + 2d - 1)^2 > 0$). Then B is decreasing in N on every range where the constants are fixed, so it suffices to evaluate it at the left end N_0 of a row. In interval arithmetic (`code/article/s6b_rigorous_checks.py`, `mpmath.iv`): for $d = 3$, $q = \frac{4}{5}$, the row (50; 1.851, 2.973) of Computer-assisted theorem 8.11 ($q_{\text{max}} = 0.8$) gives $B(50) < 0.0013$, hence the claim for $N \geq 50$; for $q = \frac{1}{2}$ the row (30; 1.962, 2.829) of Computer-assisted theorem 8.9 gives $B(30) < 0.0023$, hence $N \geq 30$. Computer-assisted theorem 8.14(b) covers $10 \leq N \leq 120$ ($q = \frac{4}{5}$) and $10 \leq N \leq 80$ ($q = \frac{1}{2}$). For $d = 2$ the rows $N_0 = 1000$ give $B(N) < 0.0100$ exactly for $N \geq 1\,761\,450$ ($q = \frac{4}{5}$) and $N \geq 554\,846$ ($q = \frac{1}{2}$), and Computer-assisted theorem 8.14(a) covers the stated finite ranges. $\square$

This settles the conjecture [CE, Conj 48] in three dimensions; in two dimensions it remains open for $302 \leq N < 1\,761\,450$ ($q = \frac{4}{5}$, apart from the five isolated certified values) and $201 \leq N < 554\,846$ ($q = \frac{1}{2}$). The trusted bases are those of the two ingredients: Arb with the transcription in `certlib.py` for the windows, and Python integers, the C engine and Arb for the finite ranges.

Conjecture 8.16 ([AS, Rem 7.19(ii), Section 8]; [CE, Conj 48]). (a) For $q = 1$ (where hypothesis (H) fails for every N and the PMF is not unimodal in general) every global maximiser still satisfies the laws of Computer-assisted theorem 8.13. (b) For $d = 2$, $q \in \{\frac{4}{5}, \frac{1}{2}\}$ and every $N \geq 10$, $|t^* - t_{\text{cf}}| \leq 0.0100t^*$.

Numerical observation 8.17 ([AS, Rem 5.3(iii)]). $X(\tau^* - \tau_{\text{cf}})$ appears to converge, to $K_2^\infty \approx 3.74$ and $K_3^\infty \approx -0.22$ (exact continuous-time modes, $d = 2$ up to $N = 1.7 \times 10^7$, $d = 3$ up to $N = 4096$; evaluation of the formal limit expression gives 3.7396 and -0.2217). The certified asymptotic windows (2.52, 4.65) ($d = 2$, $N \geq 10^{40}$) and $(-0.83, 1.43)$ ($d = 3$, $N \geq 10^6$) are consistent with this.

9 Certificates: trusted bases and how to re-verify them

Every computer-assisted statement above is reduced, by lemmas proved by hand, to finitely many inequalities that a program verifies. This section lists, per note, what the programs compute, what one has to trust, where the certificate files are, and how to re-run them. Python is a Python 3 environment with the packages of `requirements.txt` (numpy, scipy, mpmath, sympy, python-flint 0.9.0); `reproduce.md`, Section 8, gives the commands with recorded times. All recorded times are wall-clock times on a shared, often heavily loaded machine; for the single-threaded programs they are upper bounds for the CPU time.

Trusted bases.

- (T1) Python integers and `fractions.Fraction` (exact arithmetic).
- (T2) numpy `int64` limb arithmetic with an overflow bound (only for the fast generator of CE; no statement rests on it, since every certificate was re-derived under (T1)).
- (T3) Arb ball arithmetic through python-flint 0.9.0, together with the transcription of the formulas into the certification scripts (in particular `code/msc_rigorous_asymptotics/certlib.py`); in CE every Arb decision is required to agree with `mpmath.iv`.
- (T4) IEEE-754 double precision with round-to-nearest, used with a proved error model only in [1D, Prop 3.18] (with Arb at the flat points).

(T5) C99 unsigned `uint64_t/unsigned __int128` arithmetic as compiled by Apple clang 21 for arm64 with `-O2`, in the 366-line program `code/msc_rigorous_certified_c128/fp128.c`; its output is turned into verdicts with Python integers.

No program has been verified in a proof assistant. Floating-point arithmetic is otherwise used only to *find* candidates (brackets, guesses), never to decide.

statements	program(s)	base	output files; recorded time
<i>One dimension</i> (scripts in <code>code/msc_rigorous_1d/</code> , outputs in <code>data/msc_rigorous_1d/</code>)			
Thm 5.16 (c, κ, ρ)	03_constant_enclosure.py	T3	data/03_constants.json
Thm 5.8	31_tp2_powers.py (2nd impl. 34)	T1	data/31_tp2_powers_q*.json
Thm 5.17	12_cont_theorem.py, 13_cont_small_N.py, 37_kappa2.py	T3	data/12_*.json, 13_*.json, 37_kappa2.json
Thm 5.18	14_disc_theorem.py, 35_disc_uniformK.py, 15_disc_small_N.py, 32_disc_small_N_ext.py	T3	data/14_*.json, 35_disc_uniformK.json, 15_*.jsonl, 32_*.jsonl; script 14: 4–6 s per configuration (logs/rerun_all.log)
Thm 5.19	16_q1_theorem.py, 17_q1_small_N.py	T1, T3	data/16_*.json, 17_*.json
Thm 5.20	14, 35, 18_global.py	T3	data/18_global_*.json
Thm 5.23	19, 25_start_theorem.py, 26_start_small_N.py, 28	T3	data/25_*.jsonl (49 238 boxes), 26_*.jsonl (2499 pairs)
Prop. 3.18 of 1D	08_unimodality_q45.py, 24_no_underflow.py	T4, T3	data/08_modes_q45.jsonl
Prop. 5.13, Prop. 5.12 (checks)	36_start_failure.py, 40_examples.py	T1	data/36_*.json, 40_examples.json
<i>Spectral structure</i> (code/ <code>msc_rigorous_spectral/</code> ; <code>run_all.py</code> runs all scripts in one interpreter)			
Lemma 7.4	14_certify_epstein.py	T3	data/14_certified_epstein.json
Thms 7.9, 7.14(a),(b), Cor. 7.15	06_certify_constants.py, 09_certify_part3.py	T3	data/06_*.json, data/09_*.json
Thm 7.14(c),(d)	16_certify_d3_remainder.py	T3	data/16_*.json
Thm 7.10 (values)	19_certify_d2_limits.py	T3	data/19_*.json
Refuted 7.16	20_certify_rounded_constants.py	T3	data/20_*.json
Prop. 4.8	13_certify_residue_signs.py	T1, T3	data/13_*.json
finite- N enclosures [SP, Section 13]	07_certify_poles.py, 09	T3	data/07_*.json, data/09_*.json; the complete <code>run_all.py</code> took 68s in the re- run of Section 10
printed decimals	21_check_printed_decimals.py	T3	data/21_printed_decimals.json (77 strings)
<i>Asymptotics</i> (scripts in <code>code/msc_rigorous_asymptotics/</code> , outputs in <code>data/msc_rigorous_asymptotics/</code>)			
Prop. 8.5	50_cert_constants.py	T3	data/50_cert_constants.json
Thm 8.7	52_cert_theoremB.py	T3	data/52_cert_theoremB_d{2,3}.jsonl (220/389 blocks)
Thm 8.9	72_cert_discrete.py	T3	data/72_cert_discrete_d{2,3}.jsonl
Thm 8.11	76_cert_q_large.py	T3	data/76_cert_q*_d*.jsonl
Thm 8.13	78_cert_allq.py	T3	data/78_cert_allq_d{2,3}.jsonl (171/387 blocks)
tables	80_tables.py	—	data/80_tables.json; the serial run of 52, 72, 76, 78 and 80 finished within about three minutes
<i>Unimodality</i> (scripts in <code>code/msc_rigorous_unimodality/</code> , outputs in <code>data/msc_rigorous_unimodality/</code>)			
Thm 6.5	cert_unimodal.py (cross-check verify_certificate_code.py)	T1	data/cert_*.jsonl, summary_certificates.json; (a) 88 CPU-min, (b) 20 CPU-min
Thm 5.10	cert_thr1d.py (two engines)	T1	data/cert_thr1d_*.jsonl; 1.84 h (flint, $N \leq 300$), 0.44 h (Python, $N \leq 200$), 0.57 h (centre to corner)
Thm 6.3	cert_lc1d.py, cert_family.py	T3 (T1 cross- check)	data/cert_lc1d_*.jsonl, cert_family_F{1,2}.jsonl; 26 CPU- min for each of Thm 9.7(a),(b) of UN, Lemma 9.10 19 s
Thm 6.4 (small blocks)	cert_blocks.py	T3, T1	data/cert_blocks.jsonl (38 cases)
Thms 6.9, 6.10	cert_continuous.py, cert_pairs.py	T1	data/continuous_counterexamples.json, data/pairs_d*_N*.json
<i>Certified modes</i> (scripts in <code>code/msc_rigorous_certified/</code> , certificates in <code>data/msc_rigorous_certified/</code> ; recipe in [CE, Section 10])			
Thms 5.24, 6.13, 6.6 (T1 ranges)	generator fplimb.py, checker verify_packed.py (253 lines), verify_exact.py, fpcore.py	T1	certs/modes_*, packed_*, packedexact_*; 1539 certificates, all re-derived by the checker; longest verification 1794 s ($d = 3$, $N = 60$)

statements	program(s)	base	output files; recorded time
same, extended ranges	c128/fp128.c, c128_post.py	T5, T1	certs/c128_*.jsonl; 3824 certificates, each computed twice with identical sha256 digest; 28623s first pass, 30080s second pass; longest certificate 1501s
Lemma 7.3, Thm 8.14	mfpt_enclosure.py, closed_form.py	T1, T3	certs/mfpt_d*.jsonl, certs/closedform_*.jsonl
Thms 7.11, 7.13, 5.16 (2nd proof)	constants.py, mfpt2d_remainder.py	T3	certs/constants.json, certs/mfpt2d_remainder.json
all of the above	audit.py, c128/audit_ext.py	T1, T3	certs/AUDIT.json (127 assertions), certs/AUDIT_EXT.json (131 assertions)
<i>This document</i> Cor. 8.15	code/article/s6b_rigorous_checks.py (mpmath.iv)	T3	data/article/s6b_rigorous_checks.json; a few seconds

Separately written re-checks. Within the project several certificates were re-derived by separately written programs, on the same machine: in CE every Python-integer certificate by a second program with a different data layout and rounding scheme (and 553 in exact arithmetic, 622 bit for bit by a third program), and 43 of the 2285 certificates outside the Python-integer ranges (for the other 2242 the C program is the only witness, run twice); in UN the cone certificates by a dictionary-based implementation and Lemma 9.10 by a second Taylor-model implementation; in 1D the TP_2 test by a second implementation and the modes at $q = \frac{4}{5}$ against CE; in AS the certificate files of 52, 72 and 76 block by block with mpmath interval arithmetic (not 78). The code of these re-checks of the internal checks is not released. None of these is an external replication.

How to re-verify the main certificates.

1. The constant c : `python code/msc_rigorous_1d/03_constant_enclosure.py`.
2. The closed form in $d = 2, 3$: the certificate scripts `52_cert_theoremB.py`, `72_cert_discrete.py`, `76_cert_q_large.py`, `78_cert_allq.py` and `80_tables.py` of `code/msc_rigorous_asymptotics/` (commands in `reproduce.md`); then compare `data/msc_rigorous_asymptotics/80_tables.json` with the tables of Section 8.
3. Unimodality at $q \leq \frac{4}{5}$: `python code/msc_rigorous_unimodality/cert_family.py` and `python code/msc_rigorous_unimodality/summarize_lc1d.py` (the full `cert_lc1d.py` run takes about an hour of CPU).
4. Exact modes: `python code/msc_rigorous_certified/verify_packed.py data/msc_rigorous_certified/modes_d2_q4-5.jsonl` (Python integers only), and `python code/msc_rigorous_certified/audit.py`.
5. The spectral constants: `python code/msc_rigorous_spectral/run_all.py`.
6. This document: `python code/article/s6b_rigorous_checks.py` (Computer-assisted corollary 8.15); the sanity scripts of Section 10 are in the `code/msc_rigorous_*` folders (1255s wall-clock in total).

10 Re-run of the sanity checks

As part of this synthesis every sanity script of the five notes was run again, serially, one process at a time, with one BLAS thread and at the lowest scheduling priority, with a time limit of 840s per script (the driver and its per-script logs are not released; the status table and the progress log are `data/checks/sanity_rerun_status.tsv` and `data/checks/sanity_rerun_progress.log`). For SP the note's own driver `run_all.py` was used, which runs all 21 scripts of that note in one interpreter, including its certification scripts 06, 07, 09, 13, 14, 16, 19, 20, 21; for CE the two audit scripts re-check every numerical assertion of the theorems against the certificate files. The other certificate programs (the long runs listed in Section 9) were not re-run here; their stored outputs were reproduced by the line-by-line checks or, where stated in Section 11, not.

note	script	exit status	wall-clock (s)
1D	01_sanity_exact	0	3
1D	09_sec3_sec4_sanity	0	16
1D	11_expansion_sanity	0	2
1D	20_validate_theorems	0	28
1D	21_q1_sanity	0	2
1D	22_crosscheck_t5	0	1
1D	23_stress	0	39
1D	27_start_sanity	0	7
1D	29_start_stress	0	45
1D	30_tp2_pattern	0	43
1D	33_tp2_sanity	0	23
1D	34_tp2_second_impl	0	7
1D	36_start_failure	0	138
1D	38_qN_strict	0	5
1D	39_hand_constants	0	1
1D	40_examples	0	12
SP	run_all	0	68
SP	a01_recheck_separate_code	0	6
AS	02_check_structure	0	1
AS	12_identity_check	0	3
AS	40_check_thmA	0	4
AS	41_windows_exact	0	2
AS	51_validate_enclosures	0	1
AS	53_check_decomposition	0	1
AS	54_bruteforce_theorem51	0	3
AS	60_check_asymptotics	0	1
AS	70_check_discrete	0	4
AS	71_check_discrete_unimodal	0	0
AS	73_validate_discrete	0	4
AS	74_observe_q_large	0	1
AS	75_check_q_large	0	5
AS	77_validate_q_large	0	3
AS	79_validate_allq	0	183
AS	81_check_text_constants	0	0
AS	82_q1_parity_example	0	0
UN	verify_thr1d	0	80
UN	verify_lemmas	0	290
UN	verify_lastexit	0	102
UN	verify_lc1d	0	53
UN	verify_family	0	17
UN	summarize_lc1d	0	1
UN	recheck_headline	0	0
UN	scan_pairs1d	0	2
UN	table_mode_bounds_small	0	2
UN	verify_certificate_code	0	3
CE	check_lemmas	0	1
CE	check_packed	0	4
CE	check_logconcave	0	30
CE	check_remarks	0	1
CE	check_ext_lemmas	0	4
CE	audit	0	1
CE	audit_ext	0	2

Result. All 52 runs ended with exit status 0, and every verdict line they print reports success (ALL OK, ALL PASS, ALL CERTIFIED CHECKS OK, AUDIT ALL OK, “127 assertions, 0 failures”, “131 assertions, 0 failures”); total wall-clock time 1255s, with a one-minute load average between 15 and 236 caused by other processes. The SHA-256 digests of the 339 files in the data directories of the five notes were recorded before and after the run: 333 files are byte-identical. Six differ: four floating-point sanity checks of AS (02, 40, 54, 75) in the last digits of eigen-decomposition residuals, one (79) in its timing fields, and data/msc_rigorous_1d/30_tp2_pattern.json, which differs by added rows ($N = 21, \dots, 24$, with the same

16 negative triples as for $12 \leq N \leq 20$); no verdict changed.

11 Internal checks

Each note was read line by line and attacked numerically by code written separately from the note's scripts, within the same project and on the same machine (Section 1). The reports of these checks are not part of the repository; `notes/VERIFICATION.md`, Section 3, summarises them.

No statement labelled theorem or computer-assisted theorem was found false by either check; every refutation concerns a statement that is labelled as refuted here.

Final reading. After the synthesis the whole document was read once more, within the same project, with particular attention to the logical order and to the statements marked $\dagger$ and $\ddagger$, which the two checks had not seen; for the marked statements whose proofs are only outlined here, the proofs in the companion notes were read. The numerical content of the marked statements, the main certificates (exact modes and unimodality for small N , the closed-form windows against the certified modes, the constant c and the spectral constants), Computer-assisted corollary 8.15 and three randomly chosen lemmas (Lemmas 7.1, 7.2 and Computer-assisted lemma 7.4) were re-checked with separately written code within the same project, on the same machine (code and log not released); this is not an external replication. No statement was found false and no status had to be lowered. The verdicts of the final reading are summarised in `notes/VERIFICATION.md`, Section 3.

12 What remains open

1. The closed form in two dimensions, quantitatively. The window of Computer-assisted theorems 8.7–8.13 is a few units wide in $X(\tau^* - \tau_{\text{cf}})$, whereas the observed values converge to $K_2^\infty \approx 3.74$, $K_3^\infty \approx -0.22$; proving the limits needs $o(1)$ -accurate bounds for the pole data and lattice sums. As a consequence the 1% accuracy of the closed form in $d = 2$ is proved only on the certified finite ranges and for $N \geq 1761450$ ($q = \frac{4}{5}$), $N \geq 554846$ ($q = \frac{1}{2}$) (Computer-assisted corollary 8.15, Conjecture 8.16(b)). The sharper lattice-sum bounds of Computer-assisted corollary 7.15 have not been fed into the certificates of Section 8.

2. Discrete time close to $q = 1$. In $d = 2, 3$ nothing is proved for $N < N_1(q) \approx \pi q/(1 - q)$, where the pairing lemma fails; between $N_1(q)$ and N_* only part of the range is covered ($d = 2$, $\frac{1}{2} < q < 1$: $N_1(q) \leq N \leq 58$ outside Computer-assisted theorem 8.11; $d = 3$: $N \in \{10, 11\}$); all $N < 10$ ($d = 3$) and $N < 12$ ($d = 2$, discrete; for $q \leq \frac{1}{2}$ the certificate of AS also covers $N = 11$, with weaker constants that are not listed here, [AS, Rem 7.19(i)]). For $q = 1$ the PMF is not unimodal in general and the mode laws are Conjecture 8.16(a). In $d = 1$: the crossover regime $\frac{9}{10} < q < 1$ with $N \leq 600$ or $(1 - q)L^2 \lesssim 27 \ln L + 100$, where the mode interpolates between the laws of Computer-assisted theorems 5.20 and 5.19, is open; $N = 6$, $q = 0.828$ and $N = 9$, $q = 0.887$ are genuine exceptions to the window of Computer-assisted theorem 5.18. For $q = 1$ the constant of Computer-assisted theorem 5.19 is proved as $K = 10.3$ for every N and as $K = 0.031$ for $N \leq 600$; that $K \approx 0.03$ suffices for every N is only observed.

3. Unimodality for $\frac{4}{5} < q < 1$, all N .

Conjecture 12.1 (thresholds; [UN, Conjs 10.2, 10.4], [1D, Rem 3.17]). (i) For $d = 2$, CC, $\inf_N q^*(N) = q^*(9) \in [0.9986, 0.9987]$, and $q^*(N) = 1$ for infinitely many N and < 1 for infinitely many N . (ii) For $d \geq 3$, CC, $q^*(N) = 1$ for all N . (iii) For $d = 1$, end to end, $q^*(N) = \frac{2N-4}{2N-3}$ for all $N \geq 22$. (iv) For $d = 1$, centre to corner, $q^*(N) = \frac{N-3}{N-1}$ for all odd $N \geq 21$. (v) The one-dimensional free increment δh_q is log-concave iff $q \leq q_1(N)$, for every N (this would give CC unimodality for $q \leq q_1(N) \rightarrow 1$ in every dimension). (vi) Q^k is eventually TP_2 for every $q < 1$, uniformly in N (Conjecture 5.9).

Proved so far: (iii), (iv) on the ranges of Computer-assisted theorem 5.10, the upper bounds $q^* \leq \frac{2N-4}{2N-3}$, $\leq \frac{N-3}{N-1}$ for all N , (v) for $N \leq 501$ and for $q \leq \frac{4}{5}$ (Computer-assisted theorem 6.3), (ii) for $d = 3$, $N \leq 30$, $d = 4$, $N \leq 8$, $d = 5$, $N \leq 5$ (Computer-assisted theorem 6.5) and for $d = 3$, $q = 1$, $N \leq 60$ (Computer-assisted theorem 6.6). Any proof must use $q < 1$ quantitatively, since the statements fail at $q = 1$ in $d = 1, 2$.

4. Shape beyond unimodality. Log-concavity (monotone hazard rate) and the front property of the CC law in $d \geq 2$, at most two local maxima for general pairs, and strict unimodality for all N are Conjecture 6.12; in $d = 1$ log-concavity is proved for $q \leq \frac{2}{3}$ and at $q = \frac{4}{5}, \frac{17}{20}, \frac{9}{10}$ (with the exact sets of Computer-assisted theorem 5.8).

5. General starts and other geometries. In $d = 1$: explicit finite- N laws for starts with $\xi_N > \frac{1}{2}$ (they should be formulated in terms of d_0), discrete time with $q > \frac{1}{2}$, and the exact threshold $q^*(N, x_0)$ for $x_0 \geq 2$ (known: $q_N^l \leq q^* \leq q_{\text{fail}}(N - x_0)$, and q_{fail} is not sharp). In $d \geq 2$ the mode asymptotics are proved only for CC; for C2M and M2C only unimodality and the location bounds of Corollary 6.2 are proved, and the lattice sums of Section 7 were evaluated only for opposite corners.

6. Spectral structure. In $d = 2$ the monotonicity of the remainders in N , the ranges $[0.8512, 0.8831]$, $[-0.5302, -0.3508]$ for all N and the rate N^{-2} are only observed; in $d = 3$ the proved windows $[-17.4, 19.4]$, $[-82.4, 80.4]$ are far from the observed $[1.61, 1.76]$, $[-0.63, -0.29]$ and no rate is proved. Beyond Theorem 4.6 no closed description of the residue sign pattern is known.

7. Verification scope. The provenance markers do not change the stated mathematical hypotheses. Certificate 78 of Computer-assisted theorem 8.13 was regenerated: all 171 certified blocks for $d = 2$ and 387 for $d = 3$, including the terminal unbounded blocks, agree with the released records apart from execution times. The required conditions $q < 1$, $N \geq \max(N_*, N_1(q))$ and $\sin(\pi/N) \leq (1-q)/q$ remain essential. This is a rerun of the released implementation, not an independent implementation. For 2242 of the 3824 certificates of the C engine the compiled C program is the only witness (each computed twice). No program has been formally verified. Classical results are cited, not re-proved: Horn–Johnson Theorems 4.3.17, 8.3.1, 8.4.4 (numbers corroborated through secondary sources), DLMF theta and elliptic-integral identities (Lemma 9.2 of SP; not needed for the bounds), Feller’s uniqueness theorem for Laplace transforms, Jacobi’s identity and the transformation of η (Hardy–Wright Thm 357, Apostol Thm 3.1; second proof of the constant c only).

8. Material not released. The reports of the internal checks with their scripts, the scripts of the numerical search for counter-examples and the working notes are not part of the repository; they are not part of the proofs.

References

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